Prague University of Economics and Business
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Agricultural Production and Climate Change

BACHELOR THESIS

Study program: Quantitative Methods in Economics
Field of study: Statistical Methods in Economics

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Prague, May 2023

Acknowledgements

I would like to express my gratitude to my supervisor, doc. RNDr. Ivana Malá, CSc., for her patience and valuable feedback. Thanks should also go to all lecturers and teaching assistants who have accompanied me on my journey of my bachelor's studies and helped me build the foundations of my specialisation. Special thanks to my family and friends who have supported me along the way.

Abstract

Modern civilisation faces a two-fold challenge regarding food security: ensuring sufficient food supply for the growing population and tackling hunger in the poorest parts of the world. On top of that, climate change is becoming more tangible and poses a potential threat to meeting food security targets in the future. These factors motivated the investigation of agricultural and climatic indicators in this thesis. Specifically, the analysis was focused on the cases of two countries, the Netherlands and Tanzania, which were chosen as representatives of the two tails of the development ladder. The main goal of this thesis is to provide a picture of the past development of agricultural and climatic variables. The thesis can be divided into two parts. In the agricultural part, the focus is on agricultural output growth and the role of productivity. The method of agricultural growth decomposition developed by the USDA is used, and the analysis is performed on the data set of international agricultural indicators published by the USDA. In the second part, two climatic indicators are analysed: average temperatures and total precipitation. The data from ERA5 reanalysis is used for this purpose. Descriptive techniques and methods for time series modelling are used to identify the extent of the changes that occurred over the studied period. The upwards shifts in the temperatures were found in both countries, while precipitation was found to have decreased in Tanzania. The findings in this thesis provide the basis for further research on the relationships between agricultural and climatic variables.

Keywords

agriculture, TFP, productivity, climate change, time series

Abstrakt

Moderní civilizace čelí dvojí výzvě z hlediska potravinové bezpečnosti: zajištění dostatečných dodávek potravin pro rostoucí světovou populaci a řešení hladu v nejchudších částech světa. Kromě toho se stále citelněji projevuje změna klimatu, která představuje potenciální hrozbu pro splnění cílů v oblasti potravinové bezpečnosti v budoucnosti. Tyto faktory motivovaly ke zkoumání zemědělských a klimatických ukazatelů v této práci. Konkrétně se analýza zaměřila na případy dvou zemí, Nizozemska a Tanzanie, které byly vybrány jako zástupci dvou konců rozvojového žebříčku. Hlavním cílem této práce je poskytnout přehled o dosavadním vývoji zemědělských a klimatických ukazatelů. Práci lze rozdělit do dvou částí. V zemědělské části je pozornost zaměřena na růst zemědělské produkce a roli produktivity. Je použita metoda dekompozice růstu zemědělské výroby navržená Ministerstvem zemědělství USA a analýza je provedena na souboru dat mezinárodních zemědělských ukazatelů publikovaných na stránkách Ministerstva. V druhé části jsou analyzovány dva klimatické ukazatele: průměrné teploty a celkový úhrn srážek. K tomuto účelu jsou použita data z reanalýzy ERA5. K určení rozsahu změn, k nimž došlo ve zkoumaném období, jsou použity deskriptivní techniky a metody modelování časových řad. V obou zemích byly zjištěny posuny teplot směrem nahoru, zatímco v Tanzanii byl zjištěn pokles srážek. Výsledky této práce poskytují základ pro další výzkum vztahů mezi zemědělskými a klimatickými proměnnými.

Klíčová slova

zemědělství, TFP, produktivita, klimatické změny, časové řady

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List of abbreviations

UN United Nations

WPP World Population Prospects

CGIAR Consultative Group on International Agricultural Research

FAO Food and Agriculture Organisation of the United Nations

ILO International Labour Organisation

IFA International Fertilizer Association

USAID United States Agency for International Development

WUR Wageningen University & Research

IPCC Intergovernmental Panel on Climate Change

WB World Bank

PPP Purchasing power parity

SSA Sub-Saharan Africa

ASTI Agricultural Science and Technology Indicators

CAP Common Agricultural Policy

p.p. percentage points

Introduction

Food demand is projected to rise by more than 50% by 2050 (1). According to FAO (2), the main factors contributing to the rise are (i) population growth and (ii) shifts to more energy-intensive diets due to income increase. At the same time, the UN has been trying to tackle the problem of hunger and malnutrition of a large share of the global population. The second sustainable development goal has been set to achieve zero hunger by 2030 (3). However, after a steady decline between 2003 and 2014, the estimated prevalence of undernourishment[1] in the world stagnated between 2015 and 2019 and started to grow again in the following years, exacerbated by the Covid-19 pandemic. It reached 9.8% in 2021, translating into an absolute number of 767.9 million undernourished individuals worldwide (5).

Stable growth of agricultural production is one of the main factors in meeting the growing food demand and ensuring global food security. In this regard, the continuously growing productivity of agricultural production is of major importance. That is why the first part of this thesis is devoted to the analysis of the growth of agricultural production and productivity, which is measured as total factor productivity (TFP). The main goal is to analyse the past development of agricultural production and sources of its growth, that is, whether it was primarily resource-driven or productivity-driven.

Agriculture is inextricably linked to weather and climate: different crops grow best under different climatic conditions, and, especially in traditional arable agriculture, seasons define what can be planted and when. Temperature, precipitation, wind and other weather-related factors directly impact the yields. This relationship also goes in reverse: agriculture impacts the climate. It has been estimated that global food systems are accountable for a third of the global greenhouse gas emissions, which are the main cause of modern climate change, with the most considerable contribution from agriculture and land use change (6).

That is why the challenges in agriculture are currently two-fold. From one point of view, agriculture must become greener and reduce its climate footprint if the climate goals are to be met. On the other hand, it is essential to realise that modern agricultural systems were developed under the relatively stable climatic conditions of the previous centuries. The rapid unavoidable changes, which are already happening and are predicted to accelerate into the future, pose a potential threat to current practices. Hence, the second big challenge is adaptation to the new climatic conditions.

Given the increasing pressure of climate change on agriculture, the big question is the quantification of the existing and potential impacts. The implications are not straightforward: Earth's climate is a complex system with high regional and temporal variability. In some areas, warming may even lead to better agricultural outcomes, for example, due to a longer

[1] FAO estimates prevalence of undernourishment as *"the probability that a randomly selected individual from the population consumes an amount of calories that is insufficient to cover her/his energy requirement for an active and healthy life"* (4).

growing season. Therefore, it requires a detailed investigation. The first step in this analysis is to quantify the extent of the changes that have already been observed in climatic variables. In the second part of this thesis, shifts in the average temperature and precipitation over 60 years (1961-2020) are investigated. The main research question is whether the average temperatures and precipitation shifted and, if so, by how much. In addition, differences in the trends between the months are analysed.

The analysis in this thesis focuses on the specific cases of two countries: the Netherlands and Tanzania. The countries were selected as two extreme cases. The Netherlands may be viewed as an example of a success story, being one of the world's agricultural and food processing market leaders. The country is renowned for its highly efficient agriculture, innovative approaches and active research in the field. Tanzania, on the contrary, presents a case of a country in Sub-Saharan Africa (hereafter also referred to as SSA), the region unfortunately famous for its high prevalence of undernourishment and poverty, civil unrest and slow economic growth.

The stark difference between the two countries is apparent. The goal is not to compare them per se but rather look at the two sides of the coin. The example of Tanzania may help comprehend the scale of challenges the most vulnerable populations face when it comes to satisfying the most basic human needs, such as food security. On the other hand, the case of the Netherlands may provide certain insights into what the most developed agricultural systems look like nowadays and how they came to their present state.

Furthermore, contrasting these two cases highlights the tremendous gap in development between the richest and the poorest countries of the world, which is also apparent in agricultural capacities and productivity. These two systems exist simultaneously in the modern globalised world, but it is sometimes difficult to comprehend the extent of the gaps between the two tails of the development ladder. In the modern globalised economy, farmers in both countries are exposed to the global shocks coming from the volatility of the markets of agricultural commodities, global economic crises and the challenges posed by climate change. However, they have different capacities to adapt their production, different access to modern technology and information, different levels of institutions and policy frameworks, and different social securities. This is inevitably reflected in different figures from a macroeconomic perspective.

The thesis is organised into five chapters, including an introduction and conclusion. Chapter 1 briefly introduces the selected countries in order to provide a better context for the core analysis. Chapter 2 is devoted to agricultural production and productivity. Chapter 3 examines the issue of climate change. The final chapter concludes.

The analysis was performed primarily in `Python` (7) using standard packages for data analysis: `pandas` (8, 9), `numpy` (10), `statsmodels` (11), `matplotlib` (12, 13) and `seaborn` (14, 15), and partially in `R` (16, 17).

1. Overview of selected countries

The Netherlands

The Netherlands is a developed Western economy. Despite owning only 0.04% of the total agricultural land in the world (18), in 2020, it accounted for 6.8% of the global value of agricultural exports (19). It could be partly explained by reexports as the Netherlands has some of Europe's largest ports, which is why *net* exports should provide a more appropriate comparison. The figure is still impressive: the Netherlands is the second largest net agricultural exporter[1], surpassed only by such an agricultural giant as Brazil. As of 2020, 54% of the country's land area was used for agricultural purposes (18). Agriculture, forestry and fishing sectors currently provide jobs to 2% of the total persons employed in the Netherlands (20).

The Dutch agricultural sector has a well-established reputation for its high efficiency (21). The hallmark of efficient Dutch agriculture is its well-developed greenhouse infrastructure (22), allowing it to be the second-largest net exporter of tomatoes after Mexico and the third-largest net exporter of cucumbers and peppers after Spain and Mexico[1] (19), despite its northern location and cloudy and rainy weather. This is a bright illustration of the fact that suitable technological solutions can be developed to overcome shortcomings of the available natural resources.

The Dutch agricultural sector underwent substantial modernisation after the Second World War, driven by extensive policy interventions as well as general modernisation trends in Western economies. The process was characterised by reducing the number of small farmers, up-scaling the farm size, land consolidation, increasing labour productivity, specialisation of the farms and extensive educational activities for the farmers. The emphasis gradually shifted from industrialising the farm work to improve performance to the efforts to curb environmental damages from traditional intensive ways of farming. Environmental implications of agricultural practices started to be recognised by policymakers as early as the 1970s (23, 24). However, unsustainable intensive use of resources continued throughout the twentieth century, leading to land degradation and losses in biodiversity (24).

The push for greater sustainability in agriculture has become more pronounced in recent years, as environmental policy is tightening and consumer demand for sustainably produced goods is rising. The Dutch have been leading the way in this area by introducing new technologies and improving existing practices throughout the whole food industry chain (21, 24). The government has implemented a package of policies aimed at making agriculture and food systems greener on the one hand (e.g. see (25, 26, 24)) and adaptation of the agricultural sector to the anticipated risks connected with climate change on the other hand (27). These activities are supported by active research and public-private partnerships (28), the most

[1] The countries were compared by the average annual value of net exports (export value − import value) during 2016-2020 reported by FAO.

prominent example of which is the Food Valley initiative with Wageningen University and Research in its heart (29).

Agricultural policy in the Netherlands cannot be viewed in isolation from the policy of the EU. The Common Agricultural Policy (CAP) is the oldest policy at the level of the EU, operating since 1962 (30). The original goal was to increase agricultural production, which was achieved through extensive protectionist measures. It has undergone several transformations since its foundation, slowly shifting towards trade liberalisation and greater emphasis on environmental policies (31, 32). Nowadays, CAP builds upon two pillars: (i) direct payments to farmers and market measures and (ii) rural development (30). The CAP's budget amounts to one-third of the EU budget (30), while 25 years ago, it was 50% (31), highlighting that safe, stable and sustainable food production has long been among the top priorities of the EU policy. In 2023, a new reform of CAP came into force (31).

Tanzania

Tanzania is a Sub-Saharan country. It is the second largest and second most populous country after Ethiopia in Eastern Africa, with 63.6 million inhabitants as of 2021 (33). It has 26 times as much land as the Netherlands (18) and 3.5 times as large a population. It used to be classified by the World Bank as a low-income country but has been moved to the lower-middle income group since 2019 (34). Tanzanian GDP per capita (in constant 2017 international dollars) has been steadily growing since 1995 at an average annual rate of 2.7%, with the only exception of 2020, when GDP fell by 1% relative to the previous year but returned to its positive growth pattern in 2021 (35). Economic growth in Tanzania was boosted by such sectors as mining, construction and tourism (36).

Despite the gradual economic growth, the level of deprivation in the Tanzanian population remains high. The poverty rate measured by the $2.15 2017 PPP threshold[2] was cut by almost half between 2000 and 2011 (a decrease from 84% to 45%) and has stagnated around the value of 45% thereafter (38). As for food security, FAO's estimates of the prevalence of undernourishment showed a decline from 33% in 2000-2002 to 23% in 2011-2013, and it has been moving around 22-23% since then (5).

The importance of agricultural growth for Tanzania is underlined by the fact that agriculture still occupies a significant role in its economy. As of 2014, the latest reported year, two out of three Tanzanian employed individuals worked in the sector of agriculture, forestry and fishing. The share decreased from the value of 82% in 2001 (20). The sector of agriculture, forestry and fishing comprised 26.7% of GDP in 2020 (39). The estimated share of the country's land area used for agriculture was 45% in 2020 (18). The majority of the population lives in rural areas (64% in 2021), although a slow urbanisation trend has been observed since 1961: the share of the rural population has decreased by 31 p.p. since then.

[2] $2.15 2017 PPP threshold is the latest revised international poverty line that has been in use by the World Bank since September 2022 (37).

It is important to note that more than half of the projected population growth by 2050 is expected to happen in the Sub-Saharan region[3]. Unless agricultural productivity is boosted, it may yield these countries highly dependable on food imports from abroad. Specifically in Tanzania, 130 million people are expected to inhabit the country in 2050, i.e. the population is expected to double (40). Coupled with the high youth dependency ratio, this poses a threat to ensuring basic needs for the Tanzanian population and meeting such targets as zero hunger, end of poverty and sufficient access to healthcare and education, not to mention the pressure on the resources and the environment.

Given the challenges, SSA has long been a focus of many international aid and development programmes, and Tanzania is no exception. For example, CGIAR (41), the largest global research and innovation network in agriculture actively working to improve climate change adaptability of food, land, and water systems in developing countries, has historically devoted much of its efforts to projects in the SSA region. The share of the budget allocated to SSA on the total CGIAR's annual expenditures has risen from 40% in 1997 to 54% in 2021 (42, 43).

In their comprehensive review of agricultural policy in three African countries, Tanzania among them, Mdee et al. (36) identify several key points. Post-independence period (1961-1985) in Tanzanian agriculture was characterised by state-led investment, subsidy programmes and collectivisation. It was followed by a period of structural adjustment programmes beginning around 1985, when the government abolished subsidies (under pressure from international lenders (44)) and more market-driven approaches to growth were employed. The trend of liberalisation continued in the twenty-first century, with the government focusing more on poverty reduction strategies after the HIPC debt relief[4]. Meanwhile, the economy was growing, and non-farm income started gaining significance for the households' livelihoods (36).

The subsidy programmes were renewed in the twenty-first century, and after the 2007-2008 food crisis, they started to be supported by the World Bank (44). Since the UN adopted sustainable development goals (SDGs) in 2015, the policy framework has been shaped by the notions of sustainable intensification of agricultural production and green revolution[5] for Africa. These concepts sound promising on paper, but there is a large discrepancy between the pledges made by the government and the capacity to implement transformation in practice. The interventions remain fragmented, mainly pushed by the donors and international organisations, and there is a lack of focus on long-term sustainable solutions (36).

[3]In the UN medium scenario, the population in SSA is projected to increase by 1 billion people, from 1.1 billion in 2021 to 2.1 billion in 2050, whereas the global population is projected to increase from 7.9 billion in 2021 to 9.7 billion in 2050, i.e. by 1.8 billion in total (40)

[4]Highly Indebted Poor Country (HIPC) was an initiative under which OECD members offered partial debt relief to the targeted countries in exchange for the adoption of policies for poverty reduction (45).

[5]The term *green revolution* refers to the pivotal change in global agriculture during the 1960s and 1970s when the yields of major grain crops skyrocketed thanks to the improved varieties of crops and intensified use of modern inputs. The Green Revolution was a success in Asian countries, where it helped counter food insecurity. However, this revolution bypassed SSA countries (46).

2. Agricultural production and productivity

Stable growth of agricultural production is one of the main factors in meeting the growing food demand and ensuring global food security. The two complementary ways of achieving production growth are (i) increasing the total amount of *inputs* (costs) and (ii) improving the *productivity* of the inputs already in use. In agriculture, there are four major input groups: land, labour, capital (e.g. farm machinery, orchard trees and livestock) and materials (e.g. energy, fertilisers and animal feed) (47).

Further expansion of inputs may be problematic in light of the pressing environmental concerns. Therefore, increasing agricultural productivity rather than inputs has been perceived as the primary tool in solving the forthcoming challenges (48). In addition to environmental issues, expanding inputs would mean higher production costs and, hence, higher agricultural prices, which would translate into higher prices of end products for consumers, whereas achieving higher production efficiency would bring the costs down. Furthermore, increasing the quantity of inputs may have a mere short-term effect on output growth, whereas improving the productivity of resources is associated with long-term gains (49).

This chapter focuses on the analysis of trends in the growth of agricultural production, which is decomposed into contributions of input growth and productivity growth. Productivity is analysed in the framework of total factor productivity, a ratio of aggregate output to aggregate input employed in production by a certain unit (e.g. industry) over a certain period (e.g. year). By taking into account the whole mix of inputs, TFP better reflects technology and efficiency gains in production and is, therefore, superior to single input productivity measures such as crop yield per hectare of land, which would reflect both efficiency growth and growth of other inputs (48, 49). The methodology for measuring agricultural TFP growth developed by the Economic Research Unit of the US Department of Agriculture (47, 50) is used in this analysis.

The international dataset of the main agricultural indicators (51) is available on the USDA website. It contains yearly time series of the estimates of agricultural output, inputs and TFP index, as well as other components necessary for agricultural growth decomposition, for 177 countries and territories, geographic regions, groups of countries defined by World Bank income classification and global estimates for the world. The relevant time series for the Netherlands and Tanzania were used from this dataset. The period covered is 1961-2020. To compile the dataset, the authors used mainly data from the FAO[1] database and augmented their estimates with data from other organisations such as ILO[2] and IFA[3] and own and other published research (50). Due to the heterogeneity of inputs and different measuring units,

[1]FAO = Food and Agriculture Organisation of the United Nations
[2]ILO = International Labour Organisation
[3]IFA = International Fertiliser Association

which leads to ambiguities in interpreting the levels of TFP, the data set is primarily designed to compare *growth rates* of agricultural TFP rather than levels (47).

The chapter is organised as follows. Section 2.1 describes the USDA methodology of decomposing agricultural production growth into factors and measuring total factor productivity growth. Section 2.2 contains the analysis of agricultural growth in the selected countries. The methodology described in section 2.1 is applied to analyse the sources of growth. Section 2.3 concludes by emphasising the capacity gap between the richest and the poorest countries, using the example of the Netherlands and Tanzania.

2.1 USDA's model of agricultural TFP growth

Before proceeding to the data analysis, a brief overview of the TFP growth model used by the USDA is provided below. More details can be found in the original paper (47) and the updates in the latest version of online documentation (50).

Let Y and X denote aggregate agricultural output and aggregate agricultural input, respectively. TFP is simply defined as their ratio: $TFP = Y/X$. Let $gr(.)$ denote the annual growth rate of a variable. The *growth rate* of TFP is measured by first-differencing the logarithmised variable, which is equivalent to applying the logarithm to the ratio of the variable at time t and $t-1$ (2.1).

$$\begin{aligned}
gr(TFP) &= ln(TFP_t) - ln(TFP_{t-1}) \\
&= ln\left(\frac{TFP_t}{TFP_{t-1}}\right) \\
&= ln\left(\frac{Y_t/X_t}{Y_{t-1}/X_{t-1}}\right) \\
&= ln\left(\frac{Y_t}{Y_{t-1}}\right) - ln\left(\frac{X_t}{X_{t-1}}\right) \\
&= gr(Y) - gr(X).
\end{aligned} \tag{2.1}$$

The log difference is a precise growth rate measure only for continuous time, but it is often used in model derivations and empirical applications as an approximation of the growth rate in discrete time. That is, the log difference multiplied by 100 is an approximation of the precise growth rate $(TFP_t/TFP_{t-1} - 1) \cdot 100\%$. It utilises a known approximation which states that $ln(1+x) \approx x$ for a small x, and, therefore, it provides a close approximation when the growth rates are low (52, App. 1). Using logarithms has the advantage of simplifying mathematical manipulations and optimisation tasks as it converts multiplicative relations into additive ones.

From 2.1, it is apparent that the change in TFP is estimated as the *residual* between the growth rate of aggregate output and the growth rate of aggregate input. That is, if, as anticipated, output grows faster than the total input employed in production (i.e. the total real cost of production), the TFP growth rate will be positive. What specific advancements it would indicate in reality depends on the context and may not be easy to disentangle. In general, positive TFP growth could reflect a whole mix of factors such as the successful acquisition of new technology, improved quality of labour (human capital gains) and natural resources, more efficient allocation of resources, gains from up-scaling the production, more favourable climatic conditions and others (47). In addition, if some of these aspects experienced degradation while others were improved, the positive sign of the TFP growth rate would mean that the advancements outweighed the losses from the deteriorating components and vice versa.

The concept of measuring total factor productivity (or multifactor productivity) growth as the difference between output and input growth rates, not only in the agricultural context but in general in the realm of the economics of growth, is also known as *Solow residual*, named after a Nobel-prize-winning (53) economist Robert Solow, who first introduced the approach of *economic growth accounting* in his paper in 1957 (54). It stems from the Cobb-Douglas production function with a technology term and the assumption of constant returns to scale:

$$Y = AL^{1-\alpha}K^{\alpha}, \tag{2.2}$$

where Y is the output produced by employing the factors labour L and capital K, A represents the technology or, in a broader sense, total factor productivity, and α and $(1-\alpha)$ are the output elasticities of capital and labour. The growth can be measured by taking the natural logarithms of both sides and first-differencing, which would yield:

$$\begin{aligned} gr(Y) &= lnY_t - lnY_{t-1} = \\ &= (lnA_t - lnA_{t-1}) + (1-\alpha)\,(lnL_t - lnL_{t-1}) + \alpha\,(lnK_t - lnK_{t-1}) = \\ &= gr(A) + (1-\alpha)gr(L) + \alpha\; gr(K). \end{aligned} \tag{2.3}$$

That is, the growth rate of the output is decomposed into the TFP growth and a weighted average of growth rates of the inputs (labour and capital), where the weights are elasticities of those inputs. By rearranging the terms so that $gr(A)$ is on the left-hand side and all the other terms on the right-hand side, it becomes clear where the term *residual* comes from. For a textbook treatment of the Solow model in continuous time, see (52, Ch. 2). For an overview of the growth accounting method, empirical applications in past research and prominent directions for improvement in measuring economic growth, see (55).

The agricultural output Y in 2.1 is measured by the USDA ERS as real gross output. It is calculated by summing up the values of production of crop commodities and animal and fishery products expressed in 2014-2016 PPP dollars (i.e. prices are fixed at average international 2014-2016 prices for each commodity)[4]. Agricultural input X in 2.1 must reflect the

[4]Note that agricultural output in this dataset does not include ornamental flowers (50), of which the Netherlands is a famous producer and exporter (56).

heterogeneity of agricultural inputs, which leads to extension 2.4. Growth of the aggregate input is obtained as a weighted average of growth rates of all l inputs weighted by their cost shares S_j, which are assumed to be equal to the output elasticities of those inputs. The weights sum up to one.

$$gr(TFP) = gr(Y) - \sum_{j}^{l} S_j gr(X_j). \tag{2.4}$$

In the methodology employed by the USDA, cost shares are allowed to vary by decade. The $l = 4$ categories of inputs are (i) land, (ii) labour, (iii) capital and (iv) materials. Agricultural land, which can be divided into rainfed cropland, irrigated cropland and permanent pasture, is quality-adjusted during aggregation to account for the different costs and productivity of these types of land. It is measured in *rainfed cropland equivalents* (i.e. the quality weight for rainfed cropland is 1, and the weights of the other two types of the land reflect their higher or lower productivity relative to the rainfed cropland). The implications of this quality adjustment can be seen in Fig. A.1.

Labour is measured as a headcount of economically active adults primarily employed in agriculture. It is not quality-adjusted due to international data availability constraints. It means that improvements in the human capital (education, skills and health of the workers) and changes in average hours worked are counted towards the growth of the TFP rather than the growing or declining real total cost of labour as an input.

Fuglie (47) suggests several useful ways to decompose agricultural output growth into factors, and this possibility is exercised in the analysis in the following section. The simplest option is given in 2.5, where agricultural output growth is decomposed into (i) contributions of changes in the quantities of individual inputs to the growth of production costs and (ii) total factor productivity, which is achieved by trivial rearrangement of terms in 2.4. The growth rates of individual inputs remain weighted by their cost shares, reflecting their contribution to the total growth of the real cost of production.

$$gr(Y) = \sum_{j}^{l} S_j gr(X_j) + gr(TFP). \tag{2.5}$$

The second option given in 2.6 decomposes agricultural output growth into (i) the contribution of the physical expansion of land (denoted as X_1), (ii) intensified use of other inputs per hectare of land, and (iii) TFP growth. Expansion of land and increased use of other inputs per hectare are hereafter also referred to as *extensification* and *intensification*. The same decomposition is possible for any other input, for example, for labour instead of land. In that case, the terms in the decomposition equation would be interpreted as the contributions of the changes in the number of workers, changes in other inputs per worker and TFP growth.

$$gr(Y) = gr(X_1) + \sum_{j=2}^{l} S_j gr\left(\frac{X_j}{X_1}\right) + gr(TFP). \tag{2.6}$$

The principal difference, as opposed to the previous decomposition method in 2.5, is that the input of interest enters calculation in absolute quantity, i.e. without its weight S_1. It allows measuring the contribution of the physical change in the quantity of the given input employed in production versus intensified use of other inputs per unit of the input of interest instead of the contributions to the growth of production costs of each input category, which are measured in decomposition in 2.5.

It is important to stress that the land is quality-adjusted by the USDA in the process of inputs aggregation, i.e. different quality weights are given to the rainfed cropland, irrigated cropland and permanent pasture. Thanks to this fact, the extensification part in Eq. 2.6 can be further decomposed into two components: (i) physical area expansion and (ii) additional gains from expanding irrigated area, as shown in 2.7:

$$\begin{aligned}
X_1 &= X_c + \beta X_p + (\gamma - 1)X_r, \\
X_a &= X_c + \beta X_p, \\
X_i &= (\gamma - 1)X_r, \\
gr(X_1) &= \omega_a gr(X_a) + \omega_i gr(X_i).
\end{aligned} \tag{2.7}$$

In 2.7, the land X_1 is divided into total (rainfed + irrigated) cropland denoted as X_c with weight 1, pasture area X_p with weight $\beta < 1$, and additional value of the area equipped for irrigation $(\gamma - 1)X_r$. The irrigated land has weight $\gamma > 1$, reflecting higher productivity compared to the rainfed cropland. Since its physical area is already accounted for in the term X_c, 1 is subtracted from its weight γ to separate the additional value of irrigation into a separate term[5]. The first two terms summed up provide the area expansion component ($X_a = X_c + \beta X_p$), and the last term provides the irrigation expansion component ($X_i = (\gamma - 1)X_r$). The area before the quality adjustment (X_c, X_p and X_r) is measured in physical hectares. After quality adjustment with the weights β and γ, the total land X_1 is expressed in rainfed cropland equivalents. The land growth rate $gr(X_1)$ is then expressed as a weighted average of the growth rates of area expansion and irrigation expansion, where the weights are the shares of the components on the total: $\omega_a = X_a/X_1$ and $\omega_i = X_i/X_1$.

[5]The quality weights β and γ differ for every entity (country/region) and are provided in the USDA's dataset. In general, $\beta < 1$ reflects that permanent pastures are less productive than rainfed cropland. Irrigated cropland, on the other hand, has weight $\gamma > 1$. For example, if $\gamma = 2.5$, a hectare of irrigated land is seen to be, on average, 2.5 times as productive as a hectare of rainfed cropland. In the decomposition of the land expansion in equation 2.7, an additional hectare of irrigated land would be counted as +1 to the first term X_a and +1.5 to the second term X_i. When measuring the growth rates, this quality adjustment attempts to account for the fact that adding a hectare of irrigated cropland leads to higher additional output compared to adding a hectare of rainfed land, and, thus, to count it in the input growth rather than residual TFP growth.

For details on data gathering, transformation and imputation of missing values in order to compile empirical estimates of all variables employed in the model for almost all countries in the world, see the source paper (47) and online documentation (50). The analysis in the following section makes use of the dataset published by the USDA Economic Research Unit (51), which contains all necessary variables to compute growth rates as well as decompose agricultural growth into factors for 177 countries, economic and geographical regions and the world as a whole.

Certain reservations regarding the TFP measure are important to keep in mind. TFP is an analytical construct that relies on the appropriateness of the assumed production function, as well as on the accuracy of the data for outputs and inputs that enter the calculation. Even though the USDA's Economic Research Unit have put commendable effort into compiling an internationally comparable dataset, it is still infused with uncertainties coming from the potential inaccuracies in the collected data or inappropriateness of the assumed production function.

Throughout the text, the *average annual growth rate* of each component is estimated as the slope coefficient in the trend line fitted with OLS on log-transformed data, following the procedure in Fuglie (47):

$$ln(Z_t) = \alpha + \beta t + \epsilon_t, \tag{2.8}$$

where Z_t denotes the component at time t (e.g. aggregate or individual input or total output), $t = 1,2,...,T$ for T time periods (for estimation of average annual growth rate by decade, $T = 10$) and ϵ_t is the error term. Intercept α does not play a crucial role in this application; it simply ensures the correct level of the line. The core interest is in the growth rate, which is estimated as $\hat{\beta} \cdot 100\%$. Using OLS rather than average first difference eliminates the noise and helps infer the long-term direction and rate of growth of the indicator of interest from the data. Growth of TFP is obtained, in line with the described methodology, as the difference between the estimated growth of total output and the estimated growth of all other components in the corresponding equation, depending on the decomposition method.

2.2 Agricultural growth

Figure 2.1 gives the first impression of the growth of the main agricultural indicators: the total agricultural output, aggregate input and TFP. Each line shows a value with respect to the base year 1961 as indices, i.e. $100 \cdot Y_t/Y_{1961}$, $t = 1961,...,2020$. The figure also contains the population growth line as a benchmark: agricultural output is expected to grow at least as fast as population. Otherwise, the country's dependence on imports for food supply would worsen. The figure highlights the differences both in the rates of growth and in the sources of growth of agricultural output.

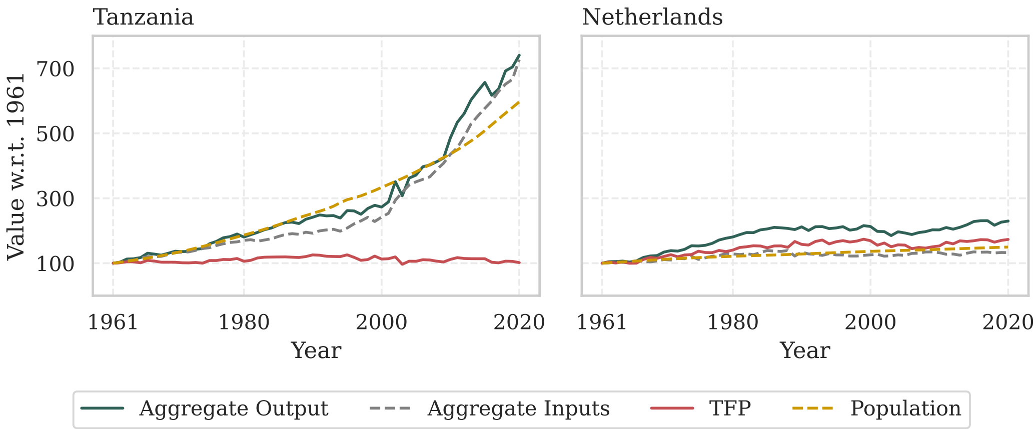

Figure 2.1: Growth of agricultural output, aggregate input, total factor productivity and growth of population in the Netherlands and Tanzania, base year 1961 = 100 (agricultural data from USDA, population data from UN WPP, Author's calculation).

In Tanzania, the growth of agricultural output was barely keeping up with the population growth in the twentieth century, but the situation seems to have improved as the agricultural growth has accelerated since the beginning of the twenty-first century. Furthermore, it is apparent that it was primarily resource-driven, while TFP seems to have stagnated for the whole period. TFP in 2020 was at practically the same level as in 1961 (index value was equal to 102, i.e. a total growth by 2%). On the contrary, the relative growth of inputs was almost identical to the output growth: output grew by 640%, while aggregate input grew by 627%). The precise growth rates will be scrutinised in more detail in the following section.

The development in the Netherlands looks rather different. The first thing to note is that the total relative growth over the 60-year period was much lower: agricultural output doubled between 1961 and 2020. However, the output grew at a much higher rate than the population. Such development is typical of high-income countries[6]. It might be explained by already relatively high production levels at the beginning of the studied period, substantially lower rates of population growth (and, hence, internal demand for agricultural products) and diminishing significance of agriculture in the overall structure of the economy. The fastest growth was observed in the first 25 years, followed by stagnation and even a slight decline during the early 2000s. Another important difference is that the TFP growth seems to have played a more important role in output growth than the increase of inputs.

The following subsections provide a more detailed look at the sources of agricultural growth in both countries with the help of decomposition methods described in section 2.1.

[6]Data for the country groups by income level are also available in the USDA's data set.

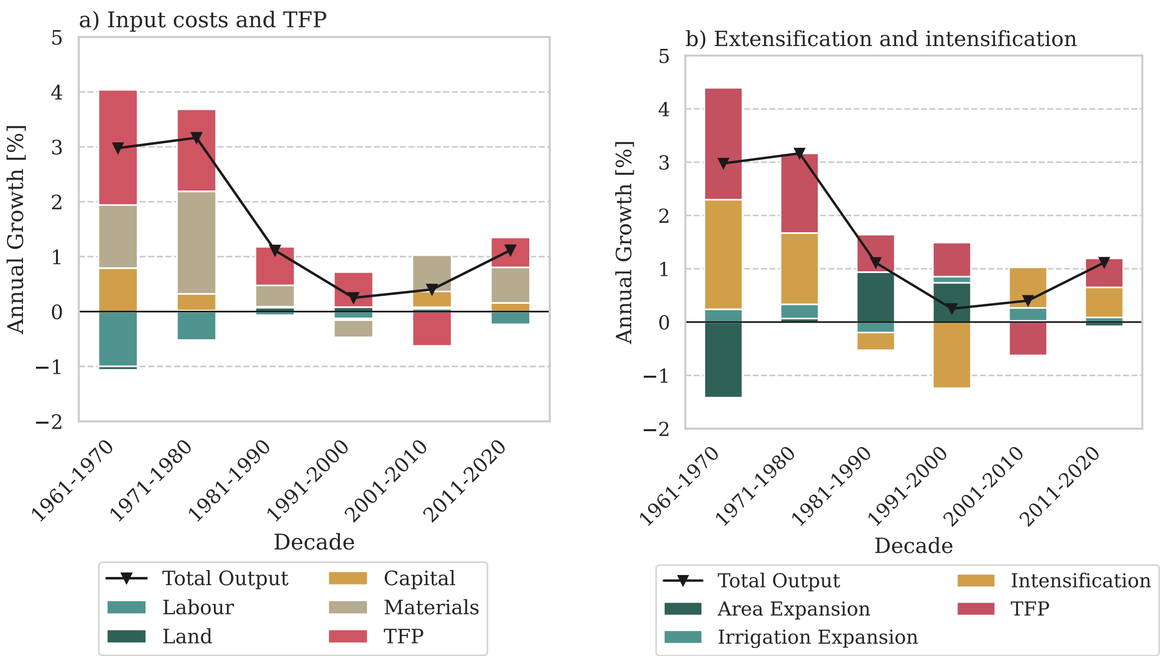

Figure 2.2: The Netherlands. Average annual growth rates of agricultural output by decade and its decomposition. a) Decomposition into contributions of inputs growth (land, labour, capital and materials) and TFP growth (Eq. 2.5). b) Decomposition into contributions of agricultural land expansion, irrigation expansion, intensification (increased use of other inputs per hectare of land) and TFP growth (Eq. 2.6 and 2.7). USDA data, Author's calculation.

2.2.1 The Netherlands

The decomposition of agricultural output growth in the Netherlands is shown in Fig. 2.2. It can be observed that the most significant changes in Dutch agriculture occurred in the first two decades of the studied period. The agricultural output grew, on average, by 3% per year. It coincided with the largest outflow of workers from agriculture and the largest gains in agricultural capital and materials. This mechanisation and rationalisation of production resulted in substantial gains in productivity: TFP growth rates were the highest during this period (average annual rates 2.1% and 1.5% during the 1960s and 1970s, respectively).

The total agricultural production in the Netherlands stagnated between the mid-1980s and 2000, followed by a slight decline in the first years of the twenty-first century, after which the slowly growing trend was reinstalled. It does not seem like a coincidence that agricultural growth peaked by the mid-1980s and then stagnated. In the 1980s, European countries ran into a problem of food production surpluses, with food supply being disproportionate to the demand. The EU policy decided to manage food supply by introducing quota on certain agricultural products (e.g. milk) since 1984 and employing other measures to restrict production (57). In the following years, other measures to phase out over-protectionism and make food production more market-oriented were employed (31), in addition to tightening environmental rules (32). These changes in the policy likely, at least partially, influenced this change in the pattern of agricultural growth in the Netherlands.

The sources of agricultural growth during these decades were different: in the 1980s and 1990s, the TFP continued to grow, although at a slower rate, while the changes in the inputs were minor, with the reduction in materials and labour being the sources of the decline during 1991-2000 (Fig. 2.2-a). Furthermore, in these two decades, area expansion played an important role in output growth, as opposed to less intensive use of other inputs per hectare (Fig. 2.2-b).

It is also important to note that a decline in TFP marked the twenty-first century's first decade. This is surprising given the high-efficiency reputation of Dutch agriculture. Negative TFP suggests a decline in the efficiency of the resource mix. The slow output growth during the 2000s was maintained due to higher capital and material use compared with the preceding decades. The situation with the productivity growth seems to have improved since 2010 as TFP returned to an upward trend and contributed to the output growth, along with higher use of materials (mainly animal feed, as the fertiliser use has been steadily decreasing).

One of the factors which may have contributed to the accelerated output growth in the last decade is the abolition of milk quotas in 2015 and the gradually relaxing rules that preceded it (58), since milk accounts for a third of the Dutch total agricultural production value and this proportion has not changed much since 1961 (59). Indeed, the data from FAO (not shown here) indicate an increasing trend in real milk production value in the last decade after a decreasing trend in the preceding years (59).

The changes in the input mix contributed to the growth of the agricultural output in the Netherlands. Labour has been steadily being replaced by capital. Between 1961 and 2020, the number of workers decreased by 65% – from 476 thousand to 166 thousand, with the fastest rates of outflow of workers in the first two decades. Meanwhile, the capital has been steadily growing, the fastest growth observed in the same decades as the outflow of labour. This corroborates the image of a highly mechanised Dutch agriculture with extensive investments into infrastructure such as greenhouses.

Furthermore, the benefits of the economies of scale have been reaped. According to the data available at Eurostat, the number of farms shrunk from 125 thousand in 1990 to 53 thousand in 2020, mainly at the expense of small and medium-sized farms, while the number of larger farms was growing (60, 61). Although the data is not available for earlier years, the literature indicates that this process began after World War II: Karel (23) mentions that the number of farms in 1945 was around 400 thousand, which would mean the closure of 275 thousand farms between 1945 and 1990. Together with the farmers' land consolidation programme and the increasing specialisation of the farms (23), it led to a more efficient, so-called *industrialised* agricultural sector.

Last but not least, successful R&D is often seen as a key ingredient in TFP growth (49), and the Dutch have proven to be a leading force in this regard (62). The sole investment into R&D is not a universal guarantee for productivity improvements, however. These new solutions must also be successfully disseminated among farmers and must be lucrative enough to be adopted. The Dutch seem to be successful in this regard, too (62, 63).

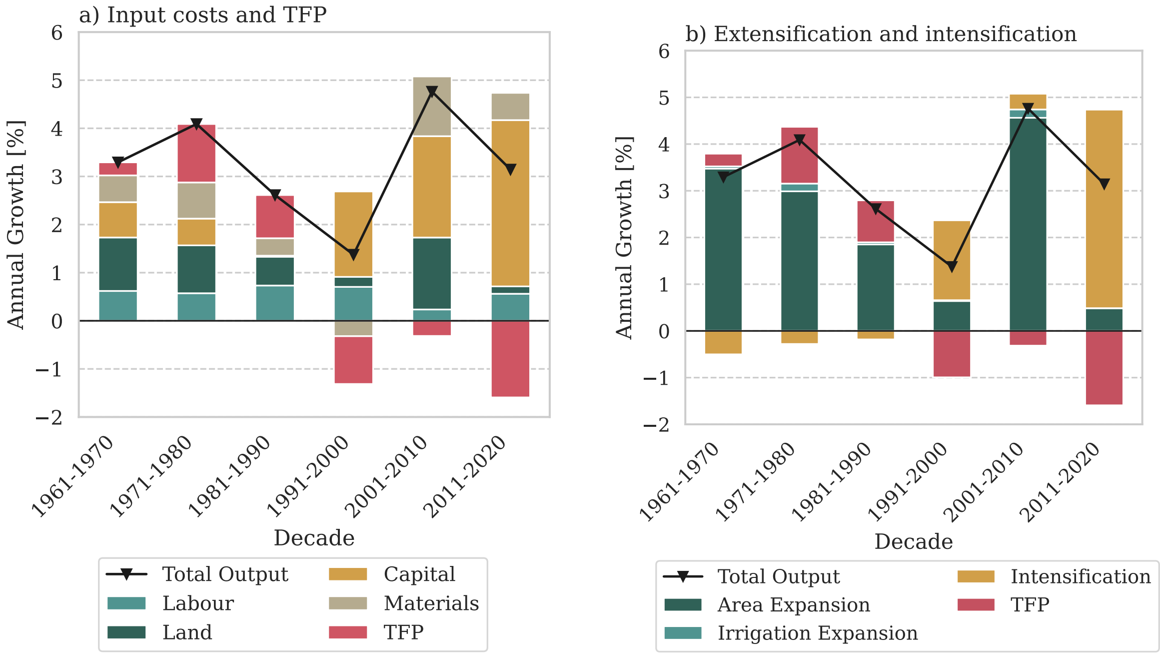

Figure 2.3: Tanzania. Average annual growth rates of agricultural output by decade and its decomposition. a) Decomposition into contributions of inputs growth (land, labour, capital and materials) and TFP growth (Eq. 2.5). b) Decomposition into contributions of agricultural land expansion, irrigation expansion, intensification (increased use of other inputs per hectare of land) and TFP growth (Eq. 2.6 and 2.7). USDA data, Author's calculation.

2.2.2 Tanzania

The decomposition of agricultural output growth in Tanzania is shown in Fig. 2.3. It can be observed that the TFP has been declining for the last three decades: TFP growth rates were estimated at -1.0%, -0.3% and -1.6% for each decade, respectively. The total agricultural output has been growing only due to the higher use of inputs. As can be seen in Fig. 2.3-a, the most notable contribution to the growth has been from the capital.

The negative TFP growth indicates that the resource mix was not used in its most efficient way, as a 1% increase in inputs (costs) led to less than a 1% increase in the output. In other words, growth in output was slower than growth in inputs (costs of production). This result is disappointing because the growth of capital and materials, which is apparent in Fig. 2.3-a, should lead to productivity gains. The example of Tanzania shows that when the whole mix of inputs is sub-optimal, this growth may not bring the anticipated benefits.

From Fig. 2.3-b, it can be observed that land expansion has played a major role in agricultural growth in Tanzania, as opposed to the intensification of other inputs and TFP. This tendency seems to have changed recently: during 2011-2020, the intensification of other inputs per unit of land was the most significant contributor to the growth. Another observation stands out in Fig. 2.3-b. The expansion of irrigation played a minor role in the total agricultural growth. This may be one of the reasons why the TFP growth was negative since water is one of the crucial resources in agriculture, and expanding irrigation infrastructure is one of the prerequisites for increasing agricultural productivity and improving food security (64, 65). When the irrigation infrastructure is underdeveloped, the harvests rely mostly on rain, which is out of the farmer's control. According to the 2019/2020 agricultural census results, 91% of the farmer households in Tanzania reported not having practised irrigation (66).

Another reason for efficiency losses may be that Tanzanian agriculture is dominated by smallholder farmers, who produce the majority of the food crops (67), meaning that family members do the majority of the work. According to the latest 2019-2020 national agricultural census in Tanzania, there were 1093 large-scale agricultural producers and 7.8 million agricultural households (representing 65% of the total number of households) (66). In Tanzania, agriculture is still a labour-intensive sector, and the number of workers has been growing throughout the studied period. On the contrary, capital and materials available per worker are extremely scarce. For example, 66.7% of households reported using neither tractors nor draft animals to cultivate the land, which implies purely human-powered work (66).

As for R&D, the report published by ASTI suggests that Tanzanian agricultural research institutions (excluding for-profit ones) have been chronically under-invested (68). This under-investment in R&D seems to be an element in a larger puzzle of limited institutional support and a lack of political will for large-scale transformation (36). The situation looks rather alarming, especially in the face of ever-growing threats posed by climate change and impressive population growth projections.

2.3 Gap in capacities

The previous section showed two different pictures of agricultural growth. Dutch agriculture underwent the most extensive transformation during the second half of the twentieth century, with minor changes from the macro perspective afterwards. It is now a typical so-called industrialised agricultural sector, which has proven highly efficient. In Tanzania, the agricultural sector is still expanding and being transformed, although the growth seems primarily resource-driven, while the efficiency seems to be declining. It is important to note that even though the agricultural output in Tanzania has a growing trend, the gap in productivity between these two countries is striking.

The following figures and conclusions in this section are based on the data from the USDA dataset – the same dataset used in the analysis in the previous section. The main figures summarising agricultural indicators' levels and growth rates are also provided in Appendix A.

As of 2020, Tanzania had 22 times the agricultural land hectarage of the Netherlands and employed 106 times more agricultural workers. With all that, Tanzanian agricultural output, as aggregated by the USDA, in constant 2015 international dollars, was 5% lower than that of the Netherlands. Tanzanian crop yield[7] has been a fraction of the Dutch crop yield since

[7] Crop yield in this context stands for the ratio of the total crop output to the total cropland area.

1961, and although certain growth was observed, the yield has not improved much. It has not yet reached even half of the yield of the Netherlands in 1961. The gap in the output per worker is even larger, reflecting the vast difference in the number of agricultural workers: the Dutch agricultural sector produced real output of 111.4 thousand constant international dollars of output per worker in 2020, against 1 thousand constant international dollars per worker in Tanzania, a difference of a magnitude more than 100, and it has stayed in a similar ratio for five decades.

These aggregate figures obviously hide the differences in the structure of agricultural sectors in these two countries: for example, livestock products account for more than two-thirds of the Dutch agricultural output, while in Tanzania, the situation is the reverse. However, a more substantive factor explaining this productivity gap between the countries is the gap in capital and modern materials employed in agriculture. The agricultural capital-to-labour ratio is very different in these two countries. For instance, in 2020, capital per worker available in the Netherlands was 365 constant international dollars, while in Tanzania, it was at 0.53 constant international dollars.

If the growth rate of capital per worker in the last decade in Tanzania is maintained, it will take more than 70 years to reach the value currently observed in the Netherlands. The estimated growth rate ($\hat{\beta} \cdot 100\%$ in equation 2.8) of capital per worker in Tanzania in 2011-2020 is 8.15% per year ($SE = 0.16$ p.p., $R^2 = 0.997$). The time needed to reach half of the value 365 under this growth rate can therefore be estimated in the following way (10 years, on which the trend was estimated, are subtracted, as $t = 1$ for the year 2011):

$$t - 10 = (ln(365/2) - \hat{\alpha})/\hat{\beta} - 10 = 81.3 - 10 = 71.3. \tag{2.9}$$

The case of these two countries is an illustration of the global phenomenon. It is hard to imagine how this gap in capacities between the richest and the poorest countries could be narrowed, but the green revolution must be achieved in Africa if the demands of the growing population are to be met. No wonder that bringing modern inputs such as fertilisers, improved varieties of crops, irrigation, and others to work on African farms is the goal of many international projects and is extensively covered in the research: for example, see (69, 70, 71, 64).

The role of institutions and policy in agricultural growth has been highlighted throughout the text, but it is important to consider one more factor, especially in terms of future prospects: climate change. All countries – rich and poor – will have to deal with it. The following chapter turns to the analysis of the climatic time series in order to quantify the changes that have already occurred since 1961 in temperatures and precipitation in the two selected countries.

3. Climate change

An impressive body of research on topics related to climate change exists nowadays, including such aspects as investigation of the climate system, its historical development, present state and possible future scenarios, as well as the extent of human impact on the climate system and the consequences it may bring for the ecosystems and humankind. This scientific evidence is regularly summarised in the Assessment Reports (AR) by the Intergovernmental Panel on Climate Change (IPCC). These reports result from the efforts of a large team of authors – e.g. more than 700 scientists contributed to AR6 – and undergo a complex multi-stage review process (72). To emphasise how extensive the body of evidence is, consider that the three parts of the latest Sixth Assessment Report (AR6) are one to three thousand pages long each.

Based on all this scientific evidence, the AR6 in 2021 unambiguously concludes: *"Human influence on the climate system is now an established fact. [...] It is unequivocal that the increase of CO_2, methane (CH_4) and nitrous oxide (N_2O) in the atmosphere over the industrial era is the result of human activities and that human influence is the main driver of many changes observed across the atmosphere, ocean, cryosphere and biosphere."* The sole fact of the warming of the Earth's climate was called *"unequivocal"* 14 years earlier, in the 2007 AR4 (73, p. 41). The IPCC working groups are bound to use the calibrated uncertainty language; hence, this phrasing is not coincidental or emotionally charged but reflects the level of confidence and agreement on the topic in the scientific community (73, Box 1.1).

An explanation of the physical basis behind the Earth's climate and climate change is beyond the scope of this text. The interested reader is referred, as a start, to a book by Andrew Dessler *Introduction to Modern Climate Change* (74), which is targeted at the non-physics audience and provides a comprehensive and accessible introduction to both physical and policy-related aspects of climate change. In short, the process of warming is the planet's response to the disturbance in its energy balance introduced by increased concentrations of greenhouse gases (GHGs) in the atmosphere as a result of intensified human economic activity since the beginning of the industrial revolution, which was powered by the combustion of fossil fuels, but other factors such as land use change played a significant role as well (74, Ch. 3–7).

Changes in the global temperature are often tracked with global temperature anomaly, defined as the difference between the average global temperature in a given year and the long-term mean of the reference period. Global temperature anomaly with respect to the average of 1951-1980 estimated by GISTEMP is shown in Fig. 3.1. The average global anomaly in the latest decade, 2011-2020, was 0.84°C[1], i.e. the global temperatures were, on average, warmer by 0.84°C compared to the level of temperatures during 1951-1980. The amount of warming has not been uniform across the globe. For example, Fig. 3.2 shows that the Northern hemisphere has been warming at a faster pace: 2011-2020 average anomaly w.r.t. 1951-1980 has been 1.09°C for the Northern hemisphere and 0.63°C for the Southern one.

[1] IPCC, which tracks changes with respect to *pre-industrial levels* (1850-1900), provides a slightly higher figure: the warming for the latest decade (2011–2020) is estimated to be 1.09°C (73, p. 59).

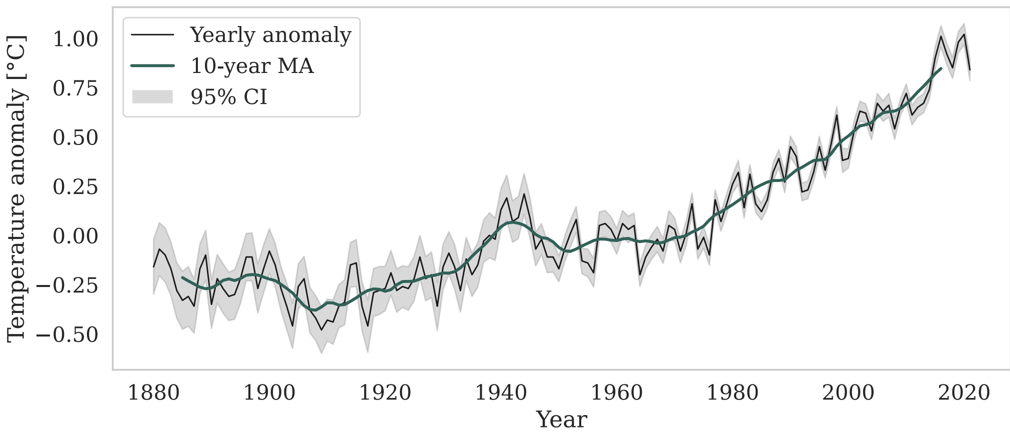

Figure 3.1: Global temperature anomaly since 1880 with respect to 1951-1980. Data including 95% CI from GISTEMP, Author's visualisation.

It may seem like a negligible value, but it certainly is not. For comparison, the difference in average global temperatures between the peaks of the ice ages and interglacials has been approximately 6°C, while the transition between these periods lasts tens of thousands of years. Reconstructions of the Earth's climate thousands of years back suggest that the average global temperature had been slowly decreasing for 7000 years before an abrupt change in trend around the 1800s when humans started adding GHGs to the atmosphere (74, Ch. 2), and the warming is not going to slow down unless the GHG emissions are cut down. Liu and Raftery (75) found that if the current trends in the growth of population, GDP per capita and CO_2 emissions per unit of GDP are sustained, the 90% prediction interval for the mean global temperature anomaly in 2100 w.r.t. 1860–1880 is [2.1°C, 3.9°C], with the median forecast 2.8°C. It would mean that roughly half of the range of the temperature difference between ice ages and interglacials would happen over two to three centuries, an unprecedented jump from the point of view of the geological time scale[2].

The sole increase in average global temperature may not seem like a big problem. After all, it is simply a calculated statistic over the entire planet and does not carry much information about local climate systems, which are of greater importance for human activities and livelihoods. However, the warming of the planet is associated with increased frequency and intensity of the local climate extremes: both wet and dry, as well as heat extremes (74, Ch. 9). The variability of precipitation is expected to increase (73, p. 82), which is critical for rainfed agriculture. Furthermore, sea level rise is one of the most certain impacts which has already been happening and is going to continue throughout the century (74, Ch. 9).

[2]According to the IPCC report, which uses scenario-based rather than probabilistic projections, the projected very likely (90% likelihood) interval for the level of warming in 2081–2100 compared to 1850–1900 is [1.0°C to 1.8°C] in the best-case scenario and [3.3°C to 5.7°C] in the worst-case scenario (73, p. 59). The best scenario (SSP1-1.9) is the path of sustainability, where CO_2 emissions peak in 2020 and start to fall gradually afterwards. The worst-case, high CO_2 emissions scenario (SSP5-8.5) is where emissions double by 2050s and peak by the end of the century (73, Infographic TS.1, p. 88–89).

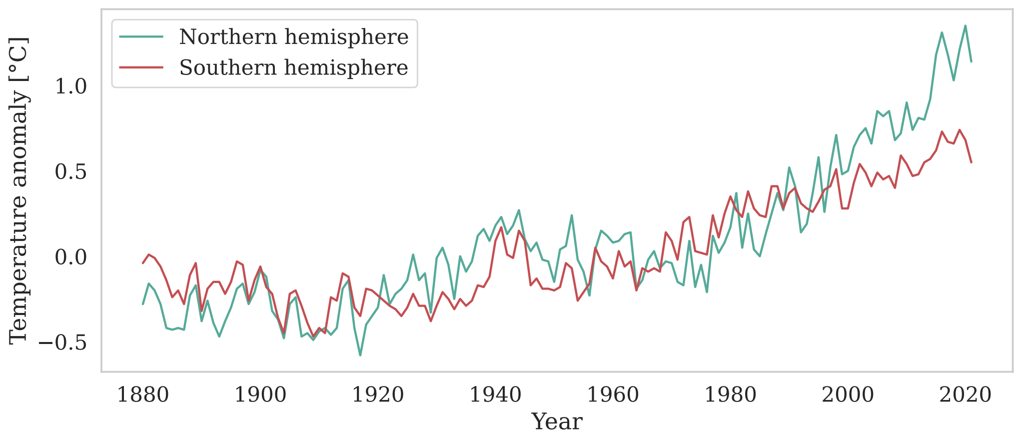

Figure 3.2: Temperature anomaly since 1880 with respect to 1951-1980 by hemisphere (GISTEMP data, Author's visualisation).

On the contrary, the growing season has increased in certain areas, which could benefit agricultural production, and elevated CO_2 concentrations may lead to increased plant growth and better water use efficiency by the plant (73, p. 82). However, there is high uncertainty about whether it can offset the negative impacts of droughts, which are expected to become more frequent (73, p. 84). The whole scope of the changes that are already occurring and are anticipated under different future scenarios are hard to comprehend and would not fit into several short paragraphs, but one thing is clear: these changes inevitably pose a threat to many human systems, which evolved under the relatively stable climate conditions of the previous centuries, agriculture among them. The larger the warming, the more severe the consequences can be expected, which is clearly shown throughout the AR6 (73).

Given the intimate relationship between human activities and climate, climate science has long ceased to be merely a niche of climate physicists. In recent years, a whole field of *climate economics* and *climate econometrics* has emerged. Econometricians use time series and panel-data techniques to provide empirical evidence of the interplay between the climate and the economy. The directions of the relationships studied are two-fold: the first group of studies focuses on the impact of economic activity on climate, whilst the second group looks at the reverse relationship – the impact of climate change on economic outcomes (76).

For example, Dell et al. (77) provide a summary of panel data studies on the relationships between weather shocks and economic outcomes, ranging from GDP to conflict and crime. For agriculture, among other findings, they highlight the importance of non-linear effects of high temperatures, with adverse effects occurring when certain thresholds in the temperatures are passed. In a later study, Letta and Tol (78) found a statistically significant negative relationship between the growth of the TFP of the whole economy and temperature shocks only in poor countries, which they attribute mainly to lowered labour productivity due to heat stress. However, these studies focus mainly on short-term annual *weather* variation over the historical record, when only limited warming has been observed. They cannot directly inform about the potential impacts of *climate* change.

In this respect, it is important to distinguish between the terms *weather* and *climate*. In general, climate refers to long-term weather patterns over a certain area (in terms of decades or longer), whereas weather is a short-term notion: month-to-month, day-to-day, hour-to-hour changes in the state of the atmosphere (74, Ch. 1). Dell et al. (77) give an intuitive analogy for statisticians and econometricians: they refer to climate as a distribution of all weather outcomes over a certain area. The observed weather is then viewed as a realisation (a random draw) from this distribution. The term *climate change* is defined by the World Meteorological Organisation (WMO) as *"a statistically significant variation in either the mean state of the climate or in its variability, persisting for an extended period (typically decades or longer)"* (79).

The rest of this chapter focuses on the climatic characteristics of the two selected countries: the Netherlands and Tanzania. More precisely, average near-surface air temperatures and total precipitation are investigated. The first two sections 3.1 and 3.2 provide a brief description of data sources and transformations and the methods used in the analysis. Sections 3.3 and 3.4 report the results for each country separately.

3.1 Data

Auffhammer et al. (80) provide a guide for economists on various available sources of climatic data, with their advantages and pitfalls. The type of climatic data used in this thesis falls under the *reanalysis* category. The data are derived from ERA5-Land reanalysis (81), produced by the European Centre for Medium-Range Weather Forecasts and distributed through Copernicus Climate Change Service (C3S). ERA5 reanalysis is a reconstruction of a wide range of climatic variables based on past observations and a complex simulation model, which uses the laws of physics to fill in the spatial and temporal gaps in observations. Citing ECMWF, *"Reanalysis data provide the most complete picture currently possible of past weather and climate"* (82).

In climate models, like ERA5, the Earth's surface and atmosphere are divided into cells, which form a grid (83). In ERA5-Land, the data are available at an *hourly frequency* for every cell, which has a horizontal latitude-longitude resolution of 0.1° x 0.1° (81). The variables used in this chapter are (i) air temperature at 2 meters above the surface of the land and (ii) total precipitation in millimetres, which represent the depth of the water that fell to the Earth's surface over a certain amount of time (in the raw data, an hour) would have if it evenly covered the bottom of the grid cell (81). The data were aggregated to the country level by applying a mask for the country area and taking a spatial average over the grid cells that fall under the mask. These procedures were performed in the CDS Toolbox (84), which allows users to retrieve and process the CDS climatic data. The online CDS Toolbox editor is a programming interface that takes `Python` code, with a wide range of useful functions available in the `cdstoolbox` library. For example, `spatial_average` function takes into account Earth's geography when calculating an average value (84). Calculations are performed on the CDS servers, and the final result is returned in the required format (e.g. a CSV file to download or an application with static or interactive charts, tables and text), which is a great advantage given the sheer volume of the data.

The code for the CDS Toolbox, which was used for data aggregation over the country territory, is provided in Appendix B. The resulting dataset included hourly time series of average temperatures and total precipitation for the Netherlands and Tanzania for the 60-year period 1961-2020. The analysis of temperature and precipitation distributions in the following sections uses *hourly* average temperature and *daily* total precipitation, which was obtained by summing up total hourly precipitation over each day. For the analysis of the monthly time series, the data were further aggregated to monthly frequency: the hourly temperature was aggregated to the average monthly temperature by taking a simple average over a given month, and the total precipitation was aggregated to average daily precipitation in a given month.

It must be noted that due to the skewness of the precipitation distribution, using the mean precipitation may not appear as the most sensible idea. It is sensible, however, to study the changes in the total amount of precipitation over a month, which is the sum of total daily precipitation. It is then practically the same as examining changes in average, which is the sum divided by the number of days in a month. Using the average eliminates the impact of different lengths of the months, which is especially important for February and, in general, for investigating differences in precipitation between the months.

3.2 Methods

The first part of the analysis investigates the shifts in climatic variables summarised over 10- and 20-year periods. The logic behind investigating statistics over a decade or longer rather than year-to-year, or even on a more granular level, is due to large internal climate variability. As is apparent, for example, from Fig. 3.1, every year can be warmer or colder than the preceding one. This annual variability has nothing to do with climate change. However, the rising trend in the temperatures does. Tracking changes in statistics calculated over longer periods, e.g. decades, allows us to eliminate the noise and follow the occurring changes better.

The following exploratory techniques were used to track changes in the distributions of temperature and precipitation:

1. Plots: histograms/kernel density plots and box plots, which show distributions of hourly temperature and daily precipitation by 20-year periods (1961-1980, 1981-2000, 2001-2020).
2. Descriptive statistics, which were calculated by decade:
   - Measures of central tendency: mean and median.
   - Other measures of position: minimum, maximum, lower and upper quartiles.

- Measures of variability: standard deviation and interquartile range (the latter is not reported but can be inferred as $\tilde{x}_{75} - \tilde{x}_{25}$, where $\tilde{x}_p$ is the p-th quantile of variable x).

The second part investigates rates of changes in indicators averaged by month to see whether any patterns can be spotted, for example, more warming in hotter months, which could potentially happen if changes in temperatures were driven primarily by more frequent extreme values in the upper tail of the distribution. Shifts in monthly temperatures and precipitation are investigated visually by plotting average anomalies over 20-year periods (1981-2000 and 2001-2020) with respect to 1961-1980 in average daily minimum, maximum and mean temperature/precipitation by month.

In addition, several time series techniques are used in this part of the analysis. Average absolute annual changes for each month m are calculated in the following way:

$$\begin{aligned} \Delta Y_{m,yr} &= Y_{m,yr} - Y_{m,yr-1}, \\ \bar{\Delta} Y_m &= \frac{\sum_{yr=1961}^{2020} \Delta Y_{m,yr}}{60}, \end{aligned} \tag{3.1}$$

where $Y_{m,yr}$ is the average climatic indicator (temperature/precipitation) in month m and year yr. It is simply an average first difference for the time series of the monthly average indicator for each month separately, i.e. 12 separate annual time series, each of length 60, are used for calculation. A positive average first difference could provide the first empirical indication of a growing trend. Furthermore, larger average absolute changes in certain seasons would signal potentially more warming as compared to other seasons.

However, these empirical indicators are inevitably influenced by the noise in the data. In order to better identify whether there were upward or downward trends in all months, linear trend functions were estimated for each month with twelve separate OLS regressions of mean monthly temperature on time:

$$Y_{m,t} = \beta_{m,0} + \beta_{m,1} t + u_{m,t}, \tag{3.2}$$

where $\{Y_{m,t}\}$ is a time series of average temperature in month m (m = 1,2,...,12) indexed by year t (t = 1,2,...,60) and $\{u_{m,t}\}$ is assumed to be a normal white noise error term (i.e. $u_{m,t} \sim NID(0, \sigma_u^2)$). The coefficients were estimated with OLS. No other functional forms for the trend were considered based on satisfactory residual diagnostics of the estimated models and visual inspection of the fitted trends against the data.

The same analysis was performed for precipitation, with the only difference of using the natural logarithm of average monthly precipitation instead of the average itself. Estimated $\hat{\beta}_{m,1}$ multiplied by 100 can then be interpreted as a relative annual growth rate. Using the log instead of the untransformed variable bounds values predicted by OLS by zero from below, which is preferred for precipitation since it cannot be negative.

It must be noted that, in practice, not all models passed the tests for normality of residuals. It has implications for the validity of the t-tests of the regression coefficients. Under the assumption of the NID error term, the t-statistics for the estimated OLS coefficients $\hat{\beta}_{m,0}, \hat{\beta}_{m,1}$ have exact small-sample Student's t-distribution (85, Ch. 16). However, even if the normality assumption is dropped, OLS estimates of coefficients in 3.2 are asymptotically normally distributed under a milder assumption of the IID error term (see (85, Ch. 16) for the proof), so the usual t-tests are asymptotically valid, with t-statistic having an asymptotic standardised normal distribution $N(0,1)$.

The assumption of IID residuals was tested by several tests, as suggested in (86, Sec. 1.6):

- ACF plots with 95% confidence intervals.
- Ljung-Box test of a joint hypothesis of no autocorrelation up to k-th lag against the alternative of serial correlation in the residuals. As an additional check of independence assumption, the same Ljung-Box test was performed on the squared residuals. The idea is that if the variables are independent, their squares are independent as well (see the so-called portmanteau tests in (86)).
- Other tests of the IID hypothesis: the turning point test and the difference-sign test. These tests test for the behaviour that would be expected of an IID sequence. For example, the difference-sign test uses the idea that the number of positive signs in a differenced IID sequence is expected to be $\frac{1}{2}(n-1)$ (86, Sec. 1.6).

All tests were performed at the 5% significance level. Of course, not rejecting the null hypothesis of zero autocorrelation or IID property is not equivalent to proving that it is true. Furthermore, the assumption of independence is much stronger than simply zero autocorrelation. If none of these tests is rejected, it may only indicate that there is insufficient evidence to conclude that the residuals are not independent. However, failing one or more of these residual diagnostic checks should draw our attention to the possible misspecification of the model.

As for the *identically distributed* part of the assumption, it is harder to test since only one observation per time point is available. Therefore, we would have to assume that the residuals are draws from the same (unknown) distribution with mean zero and variance σ_u^2. When the variables are normally distributed, equal mean and variance provide evidence of identical distribution. However, if normality is violated, identical distribution becomes a stronger assumption than simply the equality of the first two moments. To provide at least some evidence for the plausibility of this assumption, the Goldfeld-Quandt test for homoskedasticity in the residuals was performed. It tests whether the variance of the residuals from the first third of the sample is the same as the variance of the last third, while the third of observations in the centre are excluded (see (87, p.122) for a brief reference). Rejecting the null hypothesis should raise concerns about the assumption of the identical distribution of the residuals, but not rejecting it would by no means prove that they are truly identically distributed, especially when normality is violated.

In addition, normality of the residuals was checked with the Q-Q plots and the Jarque-Berra test of the hypothesis that the residuals are normally distributed against the alternative that they are not (see (87, pp. 115–116) for a reference on the Jarque-Berra test).

It must be noted that the analysis of trends in the climatic variables studied in this thesis considers them to be trend-stationary (if there is any trend) as opposed to unit-root nonstationary. It means that the existence of a deterministic trend is assumed. It simplifies the analysis, allows estimation of the trend parametrically with OLS, and the models fit the data reasonably well. Unit-root nonstationarity would imply the existence of a stochastic trend, which is an accumulation of the random shocks over time, and would require to difference the time series before modelling[3].

The twelve separate estimates of the trend slope in Eq. 3.2 may provide evidence, for example, whether upward trends were observed in the temperatures in all months. However, it is not possible to conclude whether these trends were significantly different by simply comparing the point estimates. That is why this analysis was taken one step further, and all months were modelled jointly. The idea was to compare two models: the unrestricted model, where the trend slopes were allowed to differ for all months, and the restricted model, with the same linear trend for all months.

Consider the simplest form of decomposition of time series $\{Y_t\}$ into a trend component as a function of time $T_t = f(t)$, a seasonal component S_t and a stationary residual Z_t: $Y_t = T_t + S_t + Z_t$. This approach follows the one described in (86, Sec. 1.5). If $\{Z_t\}$ is IID, the trend and seasonal components can be estimated with OLS. For example, for a monthly time series with a linear trend function, the estimated model would be as follows:

$$Y_t = \alpha + \beta t + \sum_{i=1}^{11} \gamma_i s_i + Z_t, \tag{3.3}$$

where α is intercept, β is the trend slope, γ_i are deviations of the level for months $i = 1,2,...,11$ relative to the reference (12th) month, $t = 1,2,...,T$ is time an s_i are seasonal dummies ($s_i = 1$ in month i and 0 otherwise). Z_t is and IID error term: $Z_t \sim IID(0,\sigma_u^2)$. Because the model includes an intercept, eleven dummies are included to avoid the dummy variable trap.

If, on the contrary, Z_t is not IID but follows a stationary $ARMA(p,q)$ process, model 3.3 can be framed as regression with $ARMA$ errors. It can be estimated with generalised least squares (GLS) (see (86, Sec. 6.6), (90, Sec. 3.8) and (85, Sec. 8.3)). In this text, autoregressive processes $AR(p)$ for the error term were considered:

[3]There is an interesting discussion in the literature about the nature of the trend in time series of hemispheric temperatures (see, e.g. (88, 89)). According to the unit-root nonstationarity approach, the global temperatures are considered to be cointegrated with the radiative forcing; hence, the nonstationarity is "inherited" from it. Nonstationarity of the radiative forcing is assumed to be driven by human economic activity. On the other hand, the opposing approach postulates that unit-root nonstationarity would imply that global temperatures are highly unstable, so stationary time series around a trend with a one-time break is allegedly a more appropriate approach. Various techniques are used to show that one approach is superior to the other.

$$Z_t = \phi_1 Z_{t-1} + \phi_2 Z_{t-2} + ... + \phi_p Z_{t-p} + \epsilon_t, \quad \epsilon_t \sim IID(0, \sigma_\epsilon^2), \tag{3.4}$$

where $\phi_1, \phi_2, ..., \phi_p$ are the autoregressive coefficients and $\{\epsilon_t\}$ is an IID error term sequence with mean zero and constant variance σ_ϵ^2. See, for example, (86) and (85) as a reference on ARMA models.

The estimation procedure for regression with $ARMA$ errors described in (90, p. 144) was followed. Specifically, the following steps were taken:

(i) Estimate the model with trend and seasonal components by OLS.
(ii) Identify order p of $AR(p)$ for the residuals (with the help of ACF & PACF plots).
(iii) Use GLS to re-estimate the model in (i) jointly with errors following the AR process of order p identified in the previous step; ML method of estimation was used[4].
(iv) Perform residual diagnostics of the final model: they must follow the white noise process. If there are signs of misspecification (i.e. autocorrelation in the residuals), adjust the model accordingly.

The simplest approach to modelling trend and seasonal components given in 3.3 was modified to allow the estimation of different trends for different months in the following way:

$$\begin{aligned} Y_t &= \alpha + \beta t_t^* + \sum_{i=1}^{11} \gamma_i s_{i,t} + \sum_{i=1}^{11} \delta_i s_{i,t} * t_t^* + Z_t, \\ Z_t &= \phi_1 Z_{t-1} + \phi_2 Z_{t-2} + ... + \phi_p Z_{t-p} + \epsilon_t, \quad \epsilon_t \sim IID(0, \sigma_\epsilon^2), \end{aligned} \tag{3.5}$$

where Y_t is the value of variable Y at time t, α, β, γ_i and δ_i, i = 1,2,...,11 are parameters and $s_{i,t}$ are seasonal dummy variables, with $s_{i,t} = 1$ in month i and 0 otherwise. The intercept α is the level of Y_t for the reference month (December), while the terms $\sum_{i=1}^{11} \gamma_i s_{i,t}$ allow the level of the temperatures in each month to differ as compared to December. For example, for February ($s_i = 2$) the level is $(\beta + \gamma_2)$. The time variable t_t^* is a modified trend variable, which takes on the same value for all months in the same year: e.g. for the first year in the sample 1961, t_t^* is 1 for all twelve months. It results in the following vector: $t^{*\top}$ = [1,1,1,1,1,1,1,1,1,1,1,1,2,2,2,2,2,2,2,2,2,2,2,2,3,3,...]. It affects the interpretation of β: it is interpreted as expected year-to-year change in variable Y_t, not month-to-month change, as it was in 3.3.

The reason for this modification becomes clear when the interaction terms $s_i * t_t^*$ are considered. These interaction terms allow the trend to differ for each season. For example, for January ($s_1 = 1$ and $s_i = 0$ for i = 2,3,...,11), the average year-to-year change is $(\beta + \delta_1)$. For February ($s_2 = 1$), it is $(\beta + \delta_2)$ and so on. Since December is chosen as the reference month, β is its slope coefficient. The coefficients δ_i can therefore be interpreted as increments to the average year-to-year change observed in December.

[4] See also (91) for a practical example of GLS estimation of regression with errors following an autoregressive process and likelihood-ratio tests for these models in R.

This specification allows us to test whether there are statistically significant differences in the warming between the months by testing the restriction that all δ_i in 3.5 are zero against the alternative that at least one δ_i is different from zero. The restricted model looks in the following way:

$$\begin{aligned} Y_t &= \alpha + \beta t_t^* + \sum_{i=1}^{11} \gamma_i s_{i,t} + Z_t, \\ Z_t &= \phi_1 Z_{t-1} + \phi_2 Z_{t-2} + ... + \phi_p Z_{t-p} + \epsilon_t, \quad \epsilon_t \sim IID(0, \sigma_\epsilon^2), \end{aligned} \tag{3.6}$$

It was tested with the likelihood ratio test, which compares the likelihood of the restricted model L_R and that of the unrestricted model L_U. In this case, the tested restriction is $\delta_1 = \delta_2 = ... = \delta_{11} = 0$. Under the null, the LR test statistic $LR = 2(log(L_U) - log(L_R))$ has an asymptotic chi-squared distribution with r degrees of freedom, where r is the number of restrictions (in this case, $r = 11$). See e.g. (92, pp.251-252) for a brief reference.

3.3 The Netherlands

Changes in temperature and precipitation distributions

Figure 3.3 depicts the distribution of average hourly temperatures in the Netherlands in three subsequent 20-year periods: 1961-1980, 1981-2000 and 2001-2020. It is apparent that the whole distribution has shifted slightly to the right: in addition to the centre of distribution shifting, lower temperatures are becoming less frequent, while higher temperatures occur more often.

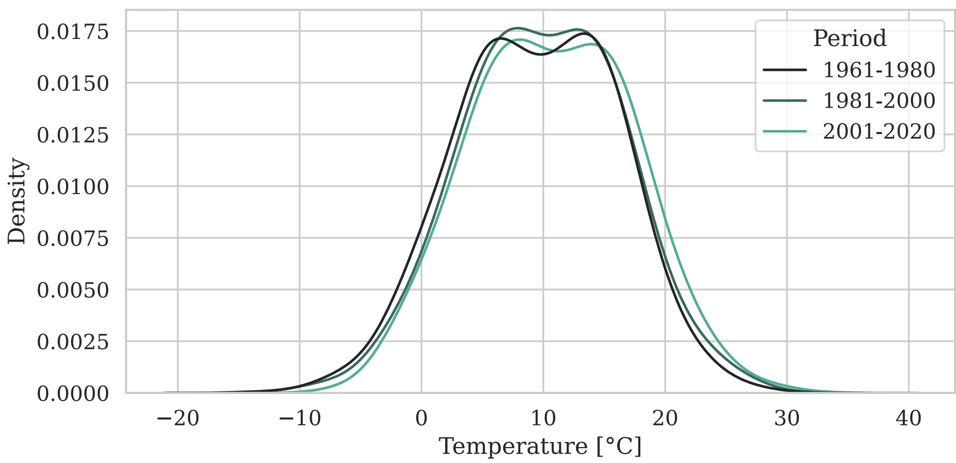

Figure 3.3: Observed values of average hourly temperatures in the Netherlands by 20-year period (ERA5 data, Author's calculation).

Decade	Mean	St.D.	Min	25%	50%	75%	Max
1961-1970	9.28	6.87	-15.70	4.32	9.70	14.51	30.80
1971-1980	9.59	6.42	-16.70	4.99	9.39	14.33	31.69
1981-1990	9.79	6.65	-16.19	5.12	9.95	14.59	32.38
1991-2000	10.14	6.64	-14.59	5.48	10.04	14.93	32.32
2001-2010	10.45	6.78	-11.47	5.49	10.55	15.48	33.52
2011-2020	10.96	6.57	-12.79	6.04	10.79	15.78	36.22

Table 3.1: Descriptive statistics of hourly average temperatures [°C] in the Netherlands by decade (ERA5 data, Author's calculation).

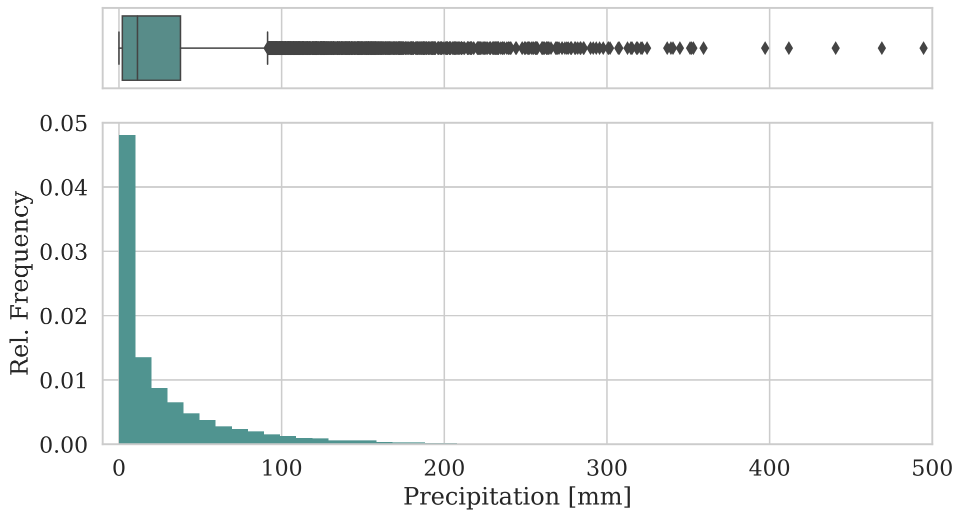

Figure 3.4: Observed values of total daily precipitation in the Netherlands over the entire period 1961-2020 (ERA5 data, Author's calculation).

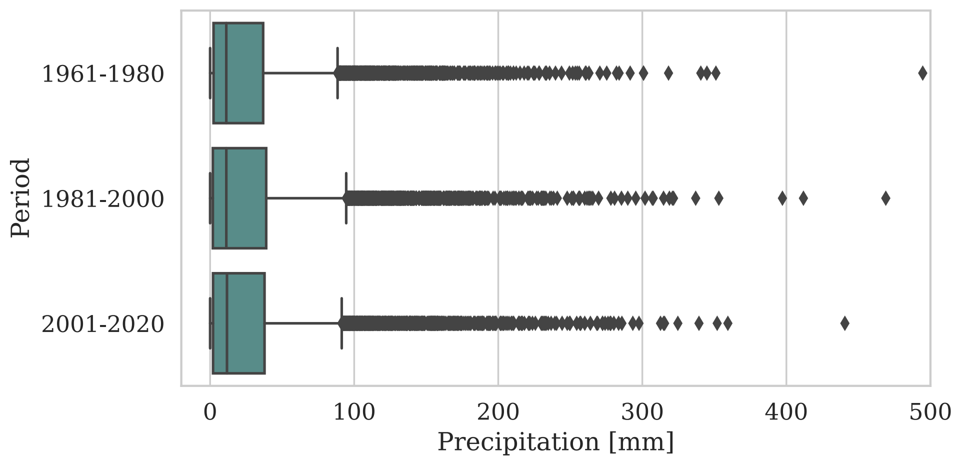

Figure 3.5: Boxplot of total daily precipitation in the Netherlands by 20-year period (ERA5 data, Author's calculation).

Decade	**Mean**	**St.D.**	**Min**	**25%**	**50%**	**75%**	**Max**
1961-1970	28.32	40.21	0.0	2.42	12.12	38.11	494.65
1971-1980	26.86	38.83	0.0	2.32	10.52	35.54	340.61
1981-1990	29.49	43.31	0.0	2.00	11.55	39.77	469.00
1991-2000	28.97	43.38	0.0	1.84	11.14	37.74	353.12
2001-2010	29.22	42.45	0.0	2.08	12.07	38.57	440.61
2011-2020	28.50	42.70	0.0	1.96	11.36	36.48	359.39

Table 3.2: Descriptive statistics of daily total precipitation [mm] in the Netherlands by decade (ERA5 data, Author's calculation).

Descriptive statistics in Tab. 3.1 corroborate the visual inspection: the gradual upward shift is apparent in the mean, as well as the median and lower and upper quartiles. The mean temperature in the latest decade, 2011-2020, was 1.68°C warmer than in 1961-1970; shifts in the quantiles were similar: 1.72°C up in the lower quartile, 1.09°C up in the median, and 1.27°C up in the upper quartile. The variability measured by the standard deviation seems relatively stable over the decades.

Precipitation distribution looks rather different (Fig. 3.4). The first thing to note is that it is substantially skewed to the right, which indicates that total precipitation on most days is in the lower range, but there is a long upper tail: days with large amounts of total precipitation also do happen relatively often. For a symmetric distribution, a common threshold for detecting the outliers in the upper tail is the value of the upper quartile plus 1.5 times the interquartile range: $\tilde{x}_{75} + 1.5(\tilde{x}_{75} - \tilde{x}_{25})$, where $\tilde{x}_p$ indicates the p-th quantile of x. This is the end of the right whisker in the boxplot. In Fig. 3.4, it is 91.4, and all the values above this threshold are depicted as dots. For such a skewed distribution, these values belong to it. In such a case, the value of 91.4 mm of precipitation per day can be interpreted as a certain upper threshold below which the body of the distribution lies.

It is also important to note that precipitation data is bound from below by zero. Fig. 3.5 indicates that there seems to have been no change in the total amount of precipitation over the three 20-year periods between 1961 and 2020. This is further corroborated by the descriptive statistics in Tab. 3.2, where there is no apparent upward or downward trend in either of them.

It may be of interest whether these changes have been evenly spread throughout the year or whether some months saw larger warming or certain changes in total precipitation, which are not apparent in the aggregated data. Fig. 3.6 shows average anomalies in average daily minimum, maximum and mean temperatures by 20-year periods for all 12 months, and Fig. 3.7 does the same for average monthly minimum, maximum and mean precipitation.

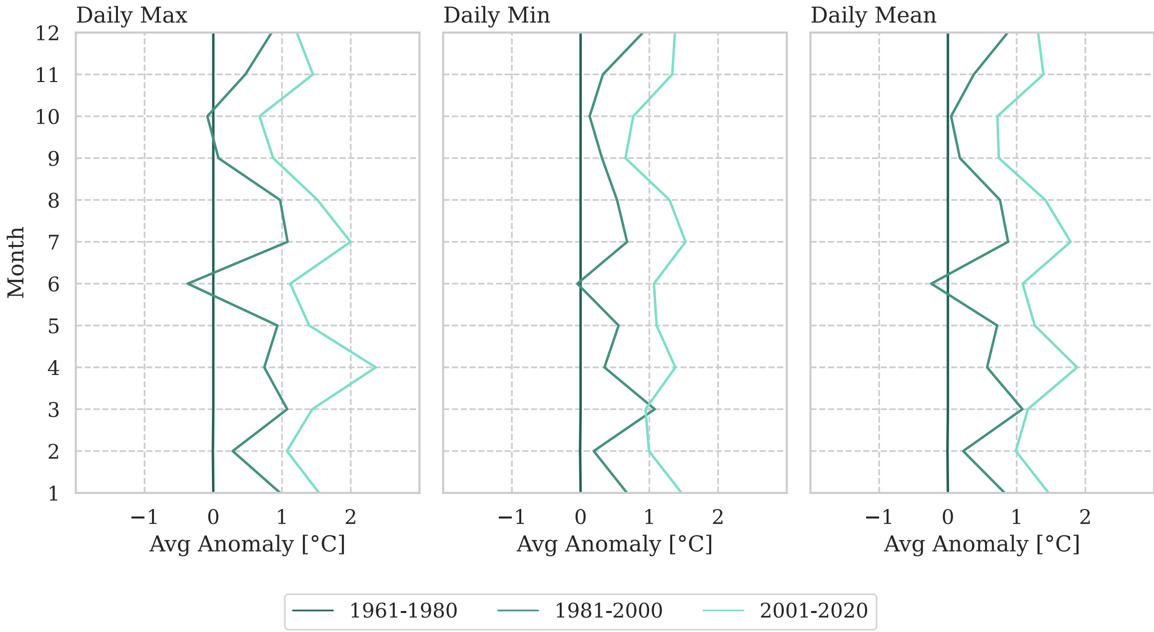

Figure 3.6: Average anomaly of average daily maximum, minimum and mean temperatures in the Netherlands w.r.t. 1961-1980 over 20-year-old periods by month. The figure shows whether the changes (if any) in temperature have been uniform across months (ERA5 data, Author's calculation)

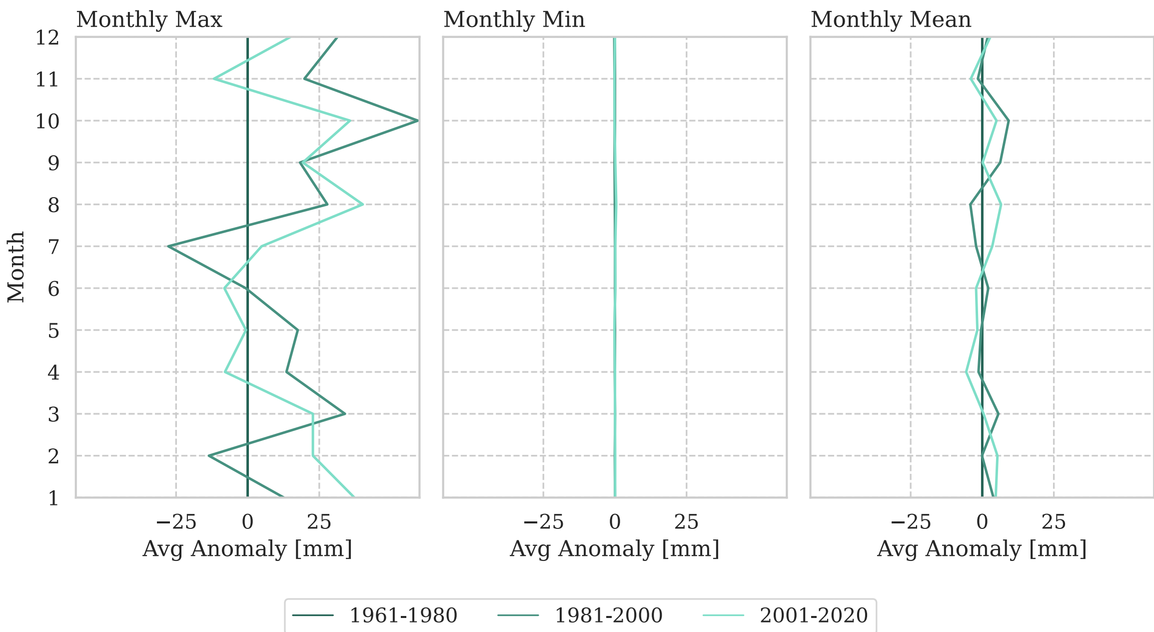

Figure 3.7: Average anomaly of maximum, minimum and mean daily precipitation by month in the Netherlands w.r.t. 1961-1980 over 20-year-old periods. The figure shows whether the changes (if any) in precipitation have been uniform across months (ERA5 data, Author's calculation)

Average anomaly for daily maximum temperature in Fig. 3.6 was calculated in the following way. Let us have a time series of daily maximum temperatures in month m, e.g. $m = 1$ (January). A yearly time series is created from it by calculating average values of daily maximum temperatures for every year y. Let it be denoted $\{Tmax_{m=1,y}\}$. The reference value for anomaly calculation is calculated as an average of $\{Tmax_{m=1,y}\}$ over $y \in$ [1961,1980]; let it be denoted $\overline{Tmax}_{m=1,ref}$. Average anomaly for a 20-year period, e.g. 1981-2000, is then calculated as $\sum_y (Tmax_{m=1,y} - \overline{Tmax}_{m=1,ref})/20$ for $y \in$ [1981,2000]. Anomalies for other months and daily minimum and mean were calculated analogously.

Anomalies for precipitation in Fig. 3.7 were calculated as an average difference between the maximum, minimum or mean daily precipitation in a given month and decade and the reference value. The reference value was the average of, respectively, the maximum, minimum or mean daily precipitation of the corresponding month over the years 1961-1980.

There do not seem to be any patterns, such as more warming in warmer months, neither for the temperature nor the precipitation. For the temperature, all months were warmer in 2001-2020 than the corresponding months in 1961-1980, e.g. the mean temperatures were, on average, between 0.72°C (October) and 1.88°C (April) warmer than their counterparts in 1961-1980. Regarding precipitation, the visual inspection does not indicate any substantial changes in either direction in any of the seasons: the observed differences, especially in the maximum values, seem to be the result of random climatic variability.

	Jan	**Feb**	**Mar**	**Apr**	**May**	**Jun**	**Jul**	**Aug**	**Sep**	**Oct**	**Nov**	**Dec**
Mean Temperature												
1961-1980	-0.12	0.08	-0.06	-0.06	-0.07	-0.09	-0.01	0.05	0.08	-0.05	-0.15	0.02
1981-2000	0.19	0.03	0.09	0.10	0.13	0.05	-0.01	0.01	0.01	0.08	0.14	0.07
2001-2020	0.09	0.06	-0.01	0.06	-0.07	0.08	0.09	0.18	-0.00	0.01	0.08	0.03
Total	**0.05**	**0.06**	**0.00**	**0.03**	**-0.00**	**0.01**	**0.02**	**0.08**	**0.03**	**0.01**	**0.02**	**0.04**
Avg. Daily Minimum												
1961-1980	-0.11	0.09	-0.04	-0.07	-0.07	-0.06	-0.01	0.04	0.07	-0.08	-0.15	-0.00
1981-2000	0.19	0.04	0.09	0.09	0.17	0.03	-0.00	-0.01	0.03	0.09	0.15	0.10
2001-2020	0.09	0.07	-0.04	-0.00	-0.12	0.08	0.04	0.17	-0.06	0.02	0.03	0.02
Total	**0.05**	**0.07**	**-0.00**	**0.01**	**-0.01**	**0.02**	**0.01**	**0.07**	**0.01**	**0.01**	**0.01**	**0.04**
Avg. Daily Maximum												
1961-1980	-0.10	0.09	-0.10	-0.07	-0.08	-0.10	-0.01	0.05	0.08	-0.06	-0.15	0.05
1981-2000	0.17	0.01	0.07	0.12	0.11	0.05	-0.01	0.04	-0.00	0.08	0.13	0.05
2001-2020	0.09	0.06	0.03	0.12	-0.04	0.08	0.10	0.19	0.04	0.01	0.12	0.03
Total	**0.06**	**0.05**	**0.00**	**0.06**	**-0.00**	**0.01**	**0.03**	**0.09**	**0.04**	**0.01**	**0.03**	**0.04**

Table 3.3: Average absolute annual change in monthly average temperature statistics in the Netherlands by month, by 20-year period and for all 60 years 1961-2020. *Note:* Average absolute annual change was calculated as average first difference for the time series of monthly average temperatures and average daily minimum and maximum observed in the months in 1961-2020 for each month separately, i.e. 12 separate annual time series for each indicator were used (ERA5 data, Author's calculation).

	Jan	Feb	Mar	Apr	May	Jun
Intercept						
Coefficient	1.568	2.193	4.607	7.441	11.791	14.795
SE	0.664	0.623	0.436	0.353	0.349	0.292
t-test (p-val.)	0.022	0.001	0.000	0.000	0.000	0.000
Trend (t)						
Coefficient	0.039	0.031	0.032	0.043	0.032	0.028
SE	0.019	0.018	0.012	0.010	0.010	0.008
t-test (p-val.)	0.042	0.088	0.013	0.000	0.002	0.001
Residuals						
St. Dev.	2.519	2.363	1.652	1.337	1.325	1.109
Skewness	-0.963	-0.477	-0.355	0.578	-0.280	0.234
Kurtosis	3.741	2.735	2.578	2.772	2.627	2.928
Residuals – tests (p-values)						
Ljung-Box (5 lags):						
Residuals	0.360	0.235	0.371	0.375	0.499	0.235
Residuals^2	0.937	0.866	0.805	0.154	0.714	0.206
Turning point	0.300	0.917	0.836	0.468	0.604	0.678
Diff-sign	0.824	0.506	0.506	0.824	0.268	0.824
Goldfeld-Quandt	0.063	0.270	0.890	0.070	0.024	0.964
Jarque-Berra	0.005	0.293	0.427	0.176	0.567	0.755
	Jul	**Aug**	**Sep**	**Oct**	**Nov**	**Dec**
Intercept						
Coefficient	16.227	16.307	14.239	10.699	5.756	2.499
SE	0.380	0.307	0.324	0.381	0.379	0.532
t-test (p-val.)	0.000	0.000	0.000	0.000	0.000	0.000
Trend (t)						
Coefficient	0.042	0.038	0.018	0.015	0.031	0.042
SE	0.011	0.009	0.009	0.011	0.011	0.015
t-test (p-val.)	0.000	0.000	0.063	0.175	0.006	0.008
Residuals						
St. Dev.	1.440	1.162	1.230	1.443	1.439	2.019
Skewness	0.649	0.350	0.544	-0.219	-0.460	-0.663
Kurtosis	3.096	2.889	3.547	2.793	3.894	3.844
Residuals – tests (p-values)						
Ljung-Box (5 lags):						
Residuals	0.309	0.094	0.379	0.001	0.277	0.398
Residuals^2	0.805	0.579	0.810	0.748	0.459	0.441
Turning point	0.836	0.078	0.917	0.001	0.178	0.038
Diff-sign	0.824	0.268	0.268	0.268	0.824	0.121
Goldfeld-Quandt	0.117	0.329	0.190	0.202	0.918	0.994
Jarque-Berra	0.120	0.534	0.157	0.746	0.128	0.046

Table 3.4: The Netherlands, monthly average temperature. Results of trend analysis, OLS regression. Sample size for each regression was 60 years. See Fig. 3.8 in Appendix for plotted time series with estimated trends. (ERA5 data, Author's calculation).

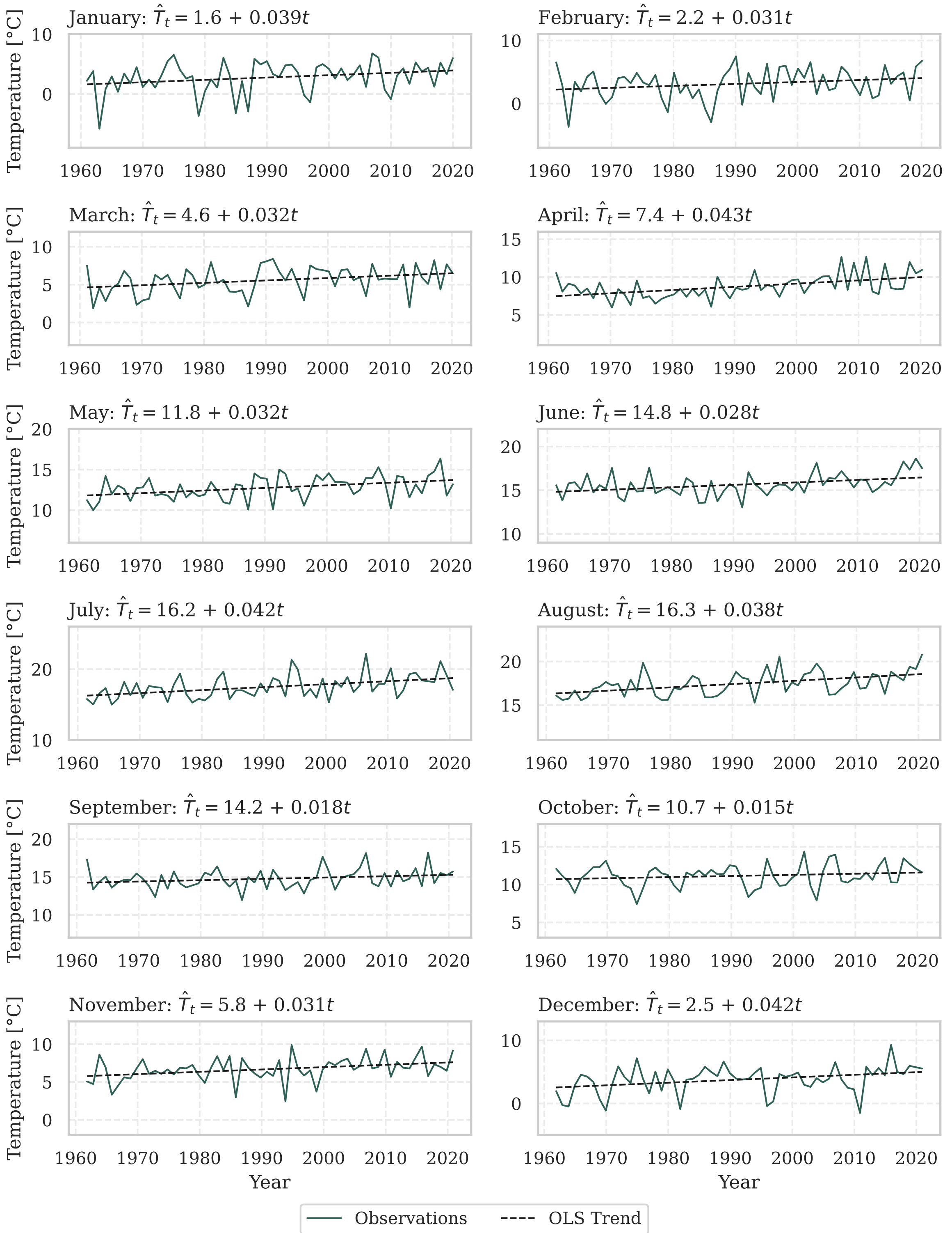

Figure 3.8: Average monthly temperatures in the Netherlands in 1961-2020 with estimated linear trends (OLS). See Tab. 3.4 for detailed OLS estimation results. *Note*: the y-axes limits are allowed to differ due to large differences in the levels of temperature between the months (ERA5 data, Author's calculation).

Trends in average monthly temperature by month

Average annual changes in temperature indicators are reported in Tab. 3.3. For example, the average year-to-year change in mean temperature between 1961 and 2020 was between 0.01°C and 0.08°C for different months, with the exception of March and May, when it was (rounded) 0.00°C. Similar patterns are observed in other indicators (average daily minimum and maximum). It does not seem like the warmer or colder months experienced substantially more or less warming, and there were even periods for certain months when the temperatures seemed, on average, to fall.

In order to better identify whether there was a positive trend in the temperatures for all months, linear trend functions were estimated for each month with twelve separate OLS regressions of mean monthly temperature on time (Eq. 3.2 in section 3.2). The results are reported in Tab. 3.4, and the estimated trends are visualised in Fig. 3.8. All estimated slope coefficients are positive and range from 0.015°C to 0.043°C of warming per year for different months. Three of them are insignificant at the 5% significance level (February, September and October). For March and May, where the empirical average annual change in mean temperature was close to zero, both estimated trend coefficients are positive and statistically significant, indicating an estimated average annual growth in mean temperatures by 0.03°C per year.

Regarding residual diagnostics, the model for October failed two tests of the IID assumption at the 5% significance level: the Ljung-Box test for no-autocorrelation in the residuals and the turning point test, which could indicate a possible misspecification for this month. For May, the Goldfeld-Quandt test rejected the hypothesis of homoskedasticity at 5% significance level. Overall, however, the models seem to be consistent with the data as the absolute majority of the tests do not reject their null hypotheses related to the IID assumption.

Joint model for average monthly temperatures

When the unrestricted and restricted models given in Eq. 3.5 and Eq. 3.6, respectively, were estimated and compared, the LR test indicated that there seemed to be no significant differences in the trends between the months (p-value 0.874). Moreover, none of the interaction terms was significant. Detailed estimation results for the restricted and unrestricted models are reported in Tab. C.1 in Appendix. OLS results are also provided for comparison (OLS was the first step in estimation, as stated in section 3.2). The differences between OLS and GLS in estimated coefficients and identification of statistically significant ones are minor. The estimated restricted model (i.e. without interaction terms) is visualised in Fig. C.1.

The temperatures were estimated to rise, on average, by 0.032°C per year, which would amount to 3.2°C of warming per century if the trend was extrapolated into the future. As discussed in the introduction to this chapter, it is an unprecedented rate of warming. The autoregressive coefficient ϕ_1 was estimated to be 0.239, indicating a weak positive autocorrelation after the deterministic trend and seasonal components are removed from the time series. It seems reasonable, as it implies that if the previous month was warmer (or colder) than predicted by the deterministic part, the subsequent month would be expected to be warmer (or colder) as well.

It must be noted that these results should be considered with caution due to the fact that the *squared* residuals of the final GLS model seem to be autocorrelated, according to the Ljung-Box test of autocorrelation in the squared residuals up to the 12th lag (see Tab. C.1). As will be shown in the following section, the same issue was detected for the temperature time series in Tanzania. This could potentially signal conditional heteroskedasticity, or, in other words, the presence of volatility clustering. It is a standard issue, for example, in financial time series. If that was indeed the case, it would clearly violate the IID assumption imposed on the error term.

The autocorrelation plot of squared residuals is shown in Fig. C.2-e. It does not look like there are any particular patterns in the autocorrelation, although lags 1, 11 and 13 do cross the 95% confidence bounds. This issue would require additional investigation into the physical properties of the air temperature and whether volatility clustering would be consistent with the theory. It could also be helpful to include additional explanatory variables in regressions 3.5 and 3.6, which would explain internal variability of the temperatures[5] and potentially eliminate the issues of autocorrelation in the (squared) residuals.

Furthermore, the residuals do not seem to be normally distributed (see Jarque-Berra test in Tab. C.1 and q-q plot in Fig. C.2-b), primarily due to outliers in the left tail. Even though non-normality is a minor issue in this case, as the sample size is sufficiently large to rely on the asymptotic results, it would be desirable to investigate the robustness of the used methods for non-normal data or extend the analysis and try to account for non-normality, for example by assuming errors from Student's t or a skewed distribution, or try robust non-parametric methods for time series modelling.

Monthly precipitation

A similar analysis for the log of average monthly precipitation did not detect any statistically significant linear trends in any of the months. This result is consistent with the initial exploratory analysis of the distribution of precipitation over 10- and 20-year periods. In addition to no trend, precipitation in the Netherlands seems to be relatively evenly distributed throughout the year, although OLS with seasonal dummies did detect statistically significantly lower average precipitation in February, March and April, at the 5% significance level, compared to the reference month (December). Furthermore, the residuals $\hat{Z}_t$ from Eq. 3.6 with a constant trend ($\beta = 0$) did not exhibit significant autocorrelation, resembling white noise (although non-Gaussian). Therefore, GLS estimation was not necessary in this case,

[5] By internal variability, it is meant that the temperatures are colder in certain years and warmer in other years, disregarding the growing trend due to climate change.

and OLS was sufficient. See Tab. C.2 in Appendix for the estimated seasonal deviations and descriptive statistics for the months and Fig. C.3 for visualisation.

3.4 Tanzania

Changes in temperature and precipitation distributions

Fig. 3.9 shows the distribution of hourly temperatures in Tanzania by 20-year periods. The shift in this figure seems even more pronounced than that in Fig. 3.3 for the Netherlands. It must be noted that in both figures, the distributions are bimodal, reflecting the diurnal temperature cycle (i.e. the fact that the nights are colder and the days are warmer). Temperatures in Tanzania also have higher levels and lower variability compared to the Netherlands. Mean temperatures over the entire 60-year period are 22.3°C and 10.0°C in Tanzania and the Netherlands, respectively. Standard deviations are 3.3°C in Tanzania and 6.7°C in the Netherlands.

The temperature statistics calculated by decade are reported in Tab. 3.5. The mean temperature observed in 2011-2020 was 0.78°C warmer than the mean temperature in 1961-1970. Variability measured by the standard deviation seems to have stayed relatively stable. The changes in the other statistics are similar: the median of the temperatures in 2011-2020 was 0.83°C higher than the median in 1961-1990; the changes in the lower and upper quartiles were 0.83°C and 0.78°C respectively. Similar to the Netherlands, the whole distribution seems to be shifting to the right.

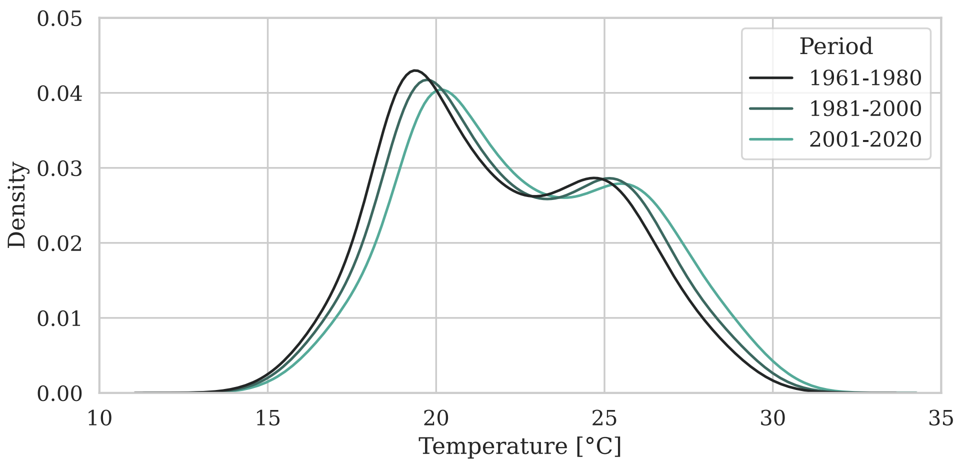

Figure 3.9: Observed values of average hourly temperatures in Tanzania by 20-year period (ERA5 data, Author's calculation).

Decade	Mean	St.D.	Min	25%	50%	75%	Max
1961-1970	21.90	3.27	13.42	19.21	21.45	24.61	31.48
1971-1980	21.87	3.20	13.22	19.29	21.43	24.49	31.25
1981-1990	22.06	3.20	13.51	19.46	21.63	24.74	31.08
1991-2000	22.39	3.36	13.62	19.68	21.96	25.17	32.06
2001-2010	22.59	3.38	13.60	19.88	22.18	25.37	32.02
2011-2020	22.68	3.29	14.00	20.04	22.28	25.39	31.61

Table 3.5: Descriptive statistics of hourly average temperatures in Tanzania by decade (ERA5 data, Author's calculation).

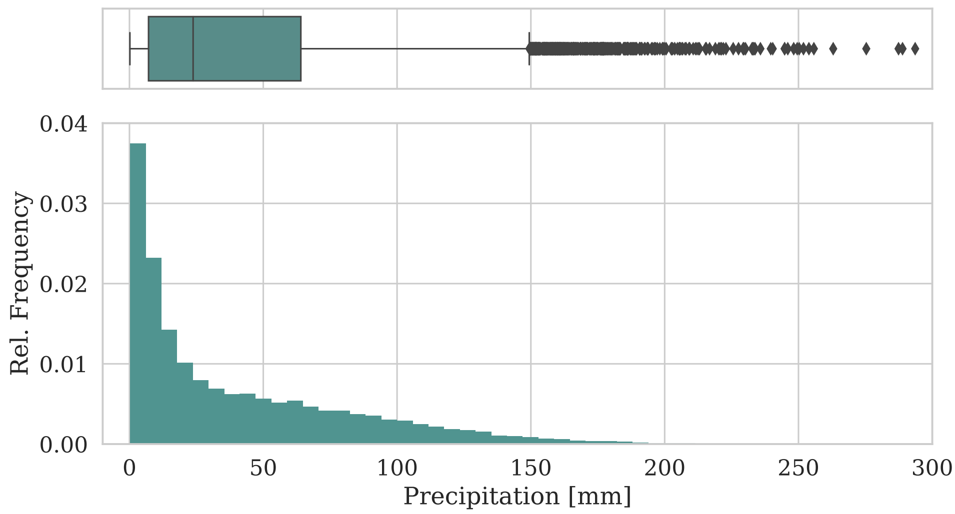

Figure 3.10: Observed values of total daily precipitation in Tanzania over the entire period 1961-2020 (ERA5 data, Author's calculation).

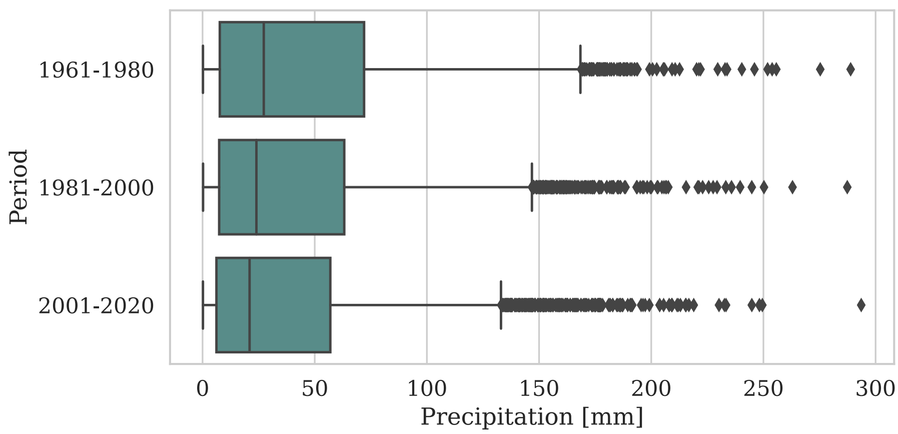

Figure 3.11: Boxplot of total daily precipitation in Tanzania by 20-year period (ERA5 data, Author's calculation).

Decade	Mean	St.D.	Min	25%	50%	75%	Max
1961-1970	43.64	44.54	0.26	7.24	25.59	71.04	288.89
1971-1980	45.33	44.56	0.18	8.10	29.14	72.96	253.90
1981-1990	43.03	41.74	0.27	7.98	28.52	69.20	262.97
1991-2000	37.20	40.59	0.23	6.69	20.42	56.85	287.39
2001-2010	34.95	38.38	0.17	5.81	18.89	52.83	249.52
2011-2020	39.46	42.09	0.17	6.63	23.22	61.22	293.60

Table 3.6: Descriptive statistics of daily total precipitation in Tanzania by decade (ERA5 data, Author's calculation).

As opposed to precipitation in the Netherlands, the total precipitation observed in Tanzania seems to be decreasing (Fig. 3.11). The overall distribution is also strongly skewed to the right, but there seems to be a heavier upper tail compared to the Netherlands. This could be due to stronger seasonal differences in precipitation in Tanzania, with heavier precipitation being more frequent in certain months. From Tab. 3.6, it is apparent that the downward shift in mean and median precipitation, as well as the quartiles, occurred in the four decades between 1971 and 2010. This decreasing tendency seems to have reversed: the latest decade, 2011-2020, saw a slight increase in precipitation.

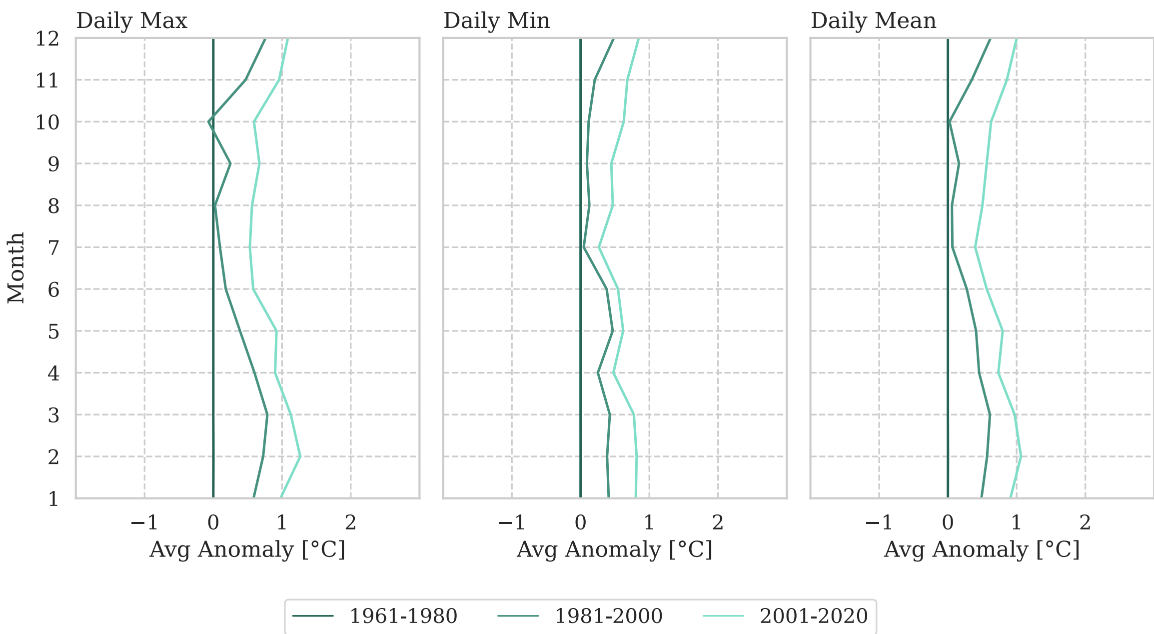

Figure 3.12: Average anomaly of average daily maximum, minimum and mean temperatures in Tanzania w.r.t. 1961-1980 over 20-year-old periods by month. The figure shows whether the changes (if any) in temperatures have been uniform across months (ERA5 data, Author's calculation)

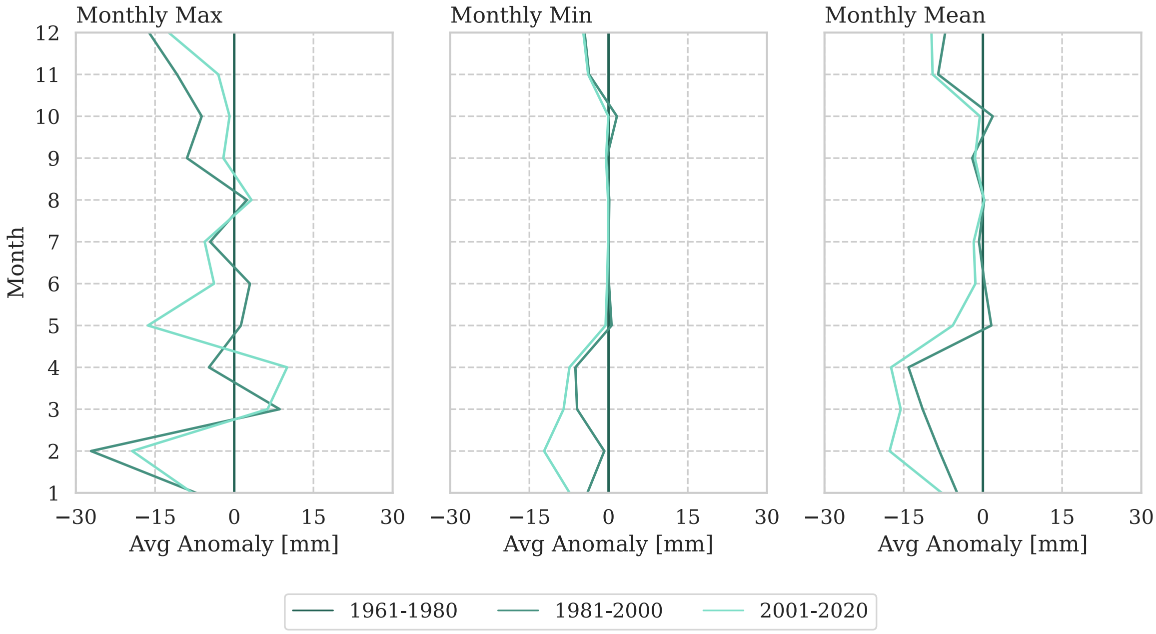

Figure 3.13: Average anomaly of maximum, minimum and mean daily precipitation by month in Tanzania w.r.t. 1961-1980 over 20-year-old periods. The figure shows whether the changes (if any) in precipitation have been uniform across months (ERA5 data, Author's calculation)

Furthermore, there seem to be certain patterns in the changes between the months, as is apparent from Fig. 3.12 and 3.13. There seems to be slightly more warming in the winter months and less warming in the summer months. As for precipitation, the shift seems to be the most pronounced between December and April. The calculation of anomalies was explained for the analogous figures for the Netherlands.

Trends in average monthly temperature by month

The same patterns, as in Fig. 3.12 and 3.13, are hard to spot in the average year-to-year changes in the temperatures and precipitation reported in Tab. 3.7 and 3.8. However, for the temperature, the same pattern occurs in the estimated OLS slope coefficients for the months (Tab. 3.9). See also Fig. 3.14 for the visual representation of the time series with the fitted trends.

The lowest estimates are for June-August (between 0.009°C and 0.014°C of warming per year) and the highest for December-March (between 0.023°C and 0.027°C). All slope coefficients are statistically significant. The models do not seem to be misspecified, according to the residual diagnostics, which are also reported in the table. One model (March) did not pass the test for homoskedasticity at the 5% significance level, while for January, the difference-sign test rejected the null hypothesis of IID residuals. Neither of the other tests for the IID assumption was rejected.

	Jan	**Feb**	**Mar**	**Apr**	**May**	**Jun**	**Jul**	**Aug**	**Sep**	**Oct**	**Nov**	**Dec**
Mean Temperature												
1961-1980	0.02	0.05	0.06	0.04	-0.00	-0.01	0.04	0.02	0.05	0.03	-0.02	-0.09
1981-2000	0.06	0.03	-0.01	0.01	0.05	0.03	-0.00	0.00	-0.00	0.02	0.02	-0.01
2001-2020	-0.03	-0.03	0.01	-0.01	-0.03	0.01	-0.00	0.02	0.01	0.00	0.01	0.05
Total	**0.02**	**0.02**	**0.02**	**0.01**	**0.00**	**0.01**	**0.01**	**0.02**	**0.02**	**0.02**	**0.00**	**-0.02**
Avg. Daily Minimum												
1961-1980	0.03	0.04	0.03	0.04	0.02	-0.04	0.03	0.03	0.03	0.04	0.02	-0.04
1981-2000	0.01	-0.00	0.00	-0.02	-0.01	0.03	-0.02	0.00	-0.02	-0.00	0.02	-0.00
2001-2020	0.03	0.03	0.03	0.03	-0.01	0.01	0.02	0.01	0.02	0.02	0.00	0.03
Total	**0.02**	**0.02**	**0.02**	**0.01**	**0.00**	**0.00**	**0.01**	**0.01**	**0.01**	**0.02**	**0.01**	**-0.00**
Avg. Daily Maximum												
1961-1980	0.01	0.07	0.09	0.05	-0.02	0.01	0.04	0.02	0.07	0.02	-0.05	-0.14
1981-2000	0.10	0.06	-0.01	0.04	0.11	0.02	0.01	0.01	0.01	0.03	0.02	-0.02
2001-2020	-0.09	-0.09	-0.03	-0.06	-0.07	-0.00	-0.02	0.03	-0.01	-0.02	0.01	0.07
Total	**0.01**	**0.01**	**0.02**	**0.01**	**0.01**	**0.01**	**0.01**	**0.02**	**0.02**	**0.01**	**-0.01**	**-0.03**

Table 3.7: Average absolute annual change in monthly average temperature statistics in Tanzania by month, by 20-year period and for all 60 years 1961-2020. *Note:* See Fig. 3.3 for explanation of average annual change calculation (ERA5 data, Author's calculation).

	Jan	**Feb**	**Mar**	**Apr**	**May**	**Jun**	**Jul**	**Aug**	**Sep**	**Oct**	**Nov**	**Dec**
Mean Daily Precipitation												
1961-1980	-0.01	-2.14	-1.91	0.10	0.90	-0.14	-0.03	0.03	-0.06	0.40	1.06	1.41
1981-2000	-2.18	0.42	0.26	-2.01	-1.13	0.15	0.00	0.03	-0.32	-0.34	0.29	0.94
2001-2020	2.61	1.94	1.59	2.03	-0.01	-0.19	0.04	-0.04	0.16	0.26	-0.21	-1.07
Total	**0.14**	**0.07**	**-0.02**	**0.04**	**-0.08**	**-0.06**	**0.00**	**0.01**	**-0.07**	**0.11**	**0.38**	**0.43**
Avg. Monthly Minimum of Daily Precipitation												
1961-1980	1.83	-1.12	-2.13	0.15	0.07	-0.03	-0.03	0.06	-0.05	-0.02	0.61	0.82
1981-2000	-1.78	-0.07	1.64	-0.35	-0.15	-0.02	-0.00	-0.03	-0.02	0.01	-0.40	-0.53
2001-2020	-0.04	0.16	0.35	0.70	-0.02	0.01	-0.01	-0.03	0.04	-0.02	-0.19	0.28
Total	**0.00**	**-0.34**	**-0.05**	**0.17**	**-0.03**	**-0.02**	**-0.01**	**0.00**	**-0.01**	**-0.01**	**0.00**	**0.19**
Avg. Monthly Maximum of Daily Precipitation												
1961-1980	-2.69	-2.62	-3.47	0.09	2.94	-0.25	-0.22	-0.13	-0.60	1.43	5.06	1.55
1981-2000	-1.51	1.99	0.05	-2.71	-3.53	1.21	0.81	0.03	-1.35	1.37	0.69	2.22
2001-2020	4.87	3.87	4.62	4.84	0.81	-1.01	0.38	-0.12	0.75	-1.63	-0.07	-1.46
Total	**0.22**	**1.08**	**0.40**	**0.74**	**0.07**	**-0.01**	**0.33**	**-0.07**	**-0.40**	**0.39**	**1.89**	**0.77**

Table 3.8: Average absolute annual change in monthly average precipitation statistics in Tanzania by month, by 20-year period and for all 60 years 1961-2020. *Note:* See Fig. 3.3 for explanation of average annual change calculation (ERA5 data, Author's calculation).

	Jan	Feb	Mar	Apr	May	Jun
Intercept						
Coefficient	21.839	21.930	21.801	21.366	21.095	20.639
SE	0.156	0.140	0.141	0.122	0.112	0.110
t-test (p-val.)	0.000	0.000	0.000	0.000	0.000	0.000
Trend (t)						
Coefficient	0.023	0.027	0.023	0.017	0.019	0.014
SE	0.004	0.004	0.004	0.003	0.003	0.003
t-test (p-val.)	0.000	0.000	0.000	0.000	0.000	0.000
Residuals						
St. Dev.	0.593	0.530	0.535	0.465	0.426	0.417
Skewness	0.498	0.422	0.599	0.488	0.176	0.134
Kurtosis	3.172	2.704	2.651	3.135	2.346	2.441
Residuals – tests (p-values)						
Ljung-Box (5 lags):						
Residuals	0.312	0.055	0.282	0.239	0.517	0.493
Residuals^2	0.669	0.997	0.881	0.932	0.597	0.711
Turning point	0.678	0.917	0.836	0.300	0.300	0.300
Diff-sign	0.046	0.824	0.824	0.506	0.824	0.268
Goldfeld-Quandt	0.927	0.318	0.011	0.353	0.080	0.167
Jarque-Berra	0.279	0.367	0.143	0.297	0.502	0.618

	Jul	Aug	Sep	Oct	Nov	Dec
Intercept						
Coefficient	20.459	21.153	22.333	23.094	22.664	21.918
SE	0.093	0.080	0.078	0.099	0.152	0.189
t-test (p-val.)	0.000	0.000	0.000	0.000	0.000	0.000
Trend (t)						
Coefficient	0.009	0.013	0.015	0.017	0.021	0.024
SE	0.003	0.002	0.002	0.003	0.004	0.005
t-test (p-val.)	0.001	0.000	0.000	0.000	0.000	0.000
Residuals						
St. Dev.	0.353	0.303	0.297	0.375	0.575	0.716
Skewness	0.297	-0.091	0.189	-0.149	-0.154	0.926
Kurtosis	2.583	3.210	2.449	2.405	2.057	3.444
Residuals - tests (p-values)						
Ljung-Box (5 lags):						
Residuals	0.765	0.212	0.077	0.427	0.376	0.210
Residuals^2	0.706	0.739	0.377	0.555	0.956	0.780
Turning point	0.917	0.604	0.604	0.407	0.917	0.300
Diff-sign	0.824	0.506	0.121	0.506	0.268	0.268
Goldfeld-Quandt	0.604	0.092	0.392	0.079	0.152	0.993
Jarque-Berra	0.517	0.908	0.572	0.575	0.292	0.011

Table 3.9: Tanzania, monthly average temperature. Results of trend analysis, OLS regression. Sample size for each regression is 60 years. See Fig. 3.14 below for the plotted time series with the estimated trends (ERA5 data, Author's calculation).

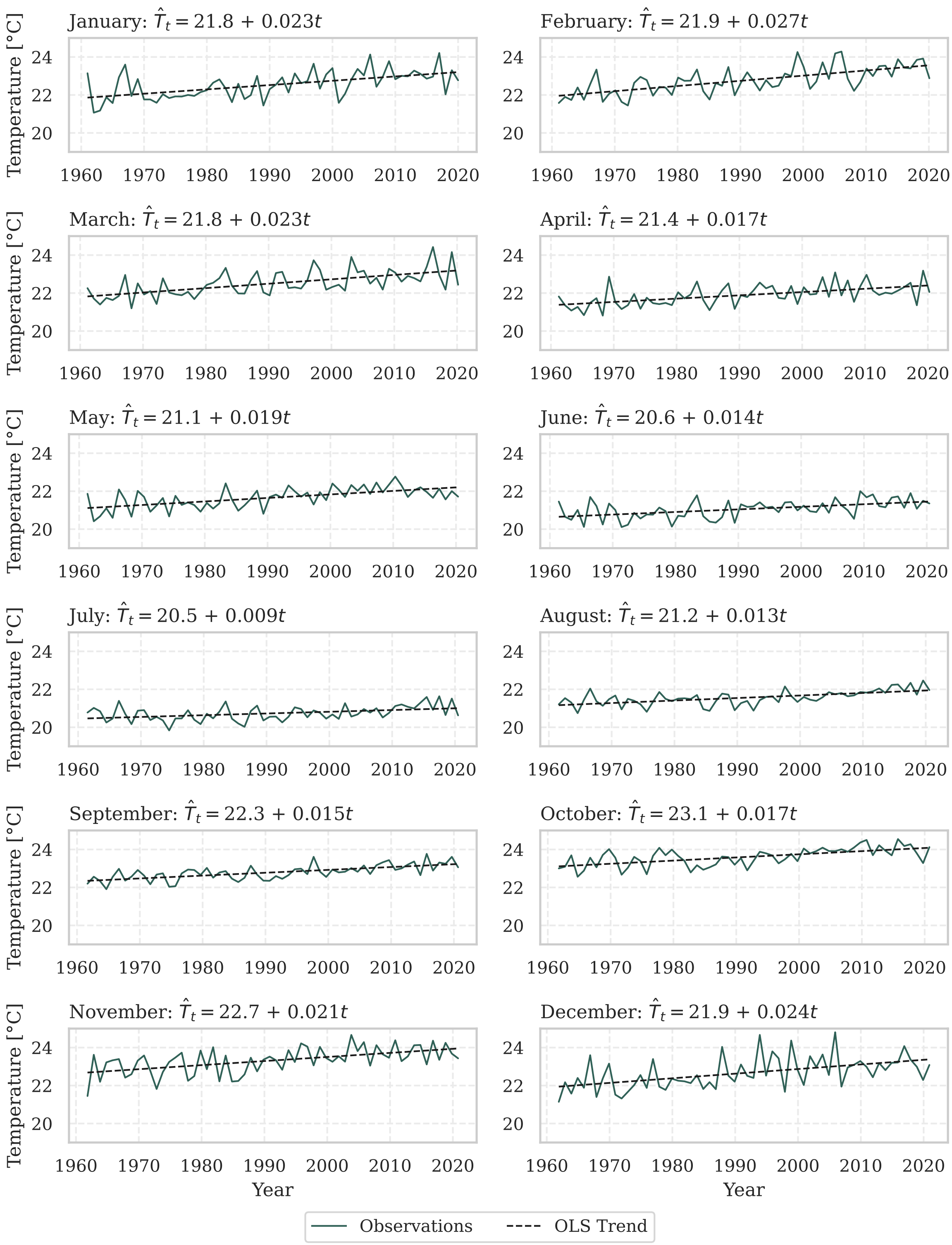

Figure 3.14: Average monthly temperatures in Tanzania in 1961-2020 with the estimated linear trends (OLS). See Tab. 3.9 above for detailed OLS estimation results (ERA5 data, Author's calculation).

Joint model for average monthly temperatures

The p-value of the LR test of the restricted model (Eq. 3.6) against the unrestricted model (Eq. 3.5) is 0.082, indicating that there is insufficient evidence to reject the null hypothesis that the temperatures in all months follow the same trend at the 5% significance level. The average warming per year was estimated to be 0.018°C by the restricted model, which is a lower rate of warming than was estimated for the Netherlands. The residuals Z_t were estimated to follow an $AR(2)$ process with coefficients $\hat{\phi}_1 = 0.45$ and $\hat{\phi}_2 = 0.12$, indicating a stronger autocorrelation than was observed in the model for the temperatures in the Netherlands. Furthermore, the seasonal deviations in temperatures in Tanzania are of much lower magnitude than in the Netherlands, which is consistent with the observation made in exploratory analysis that the spread of the temperatures is lower.

There seems to be no significant autocorrelation up to the 12th lag in the residuals of the restricted GLS model (the p-value of the Ljung-Box test is 0.19). However, the test rejects the hypothesis of no autocorrelation in the squared residuals (p-value 0.001). This is the same pattern as was observed in the model for the temperatures in the Netherlands. Inspection of the ACF plot of squared residuals (Fig. C.5) revealed a potential cyclical pattern. It is a concerning sign since it means that the IID assumption seems to be violated. Evidently, the model fails to capture all the dynamics in the temperature time series. Further investigation of this property of the temperature time series is unfortunately out of the scope of this text.

Trends in precipitation by month

As for the estimated slopes for the natural logarithm of the average monthly precipitation, all estimated slopes are negative, but they are statistically insignificant at the 5% significance level for 8 out of 12 months (Tab. 3.10). The statistically significant negative slopes were found in February, March and April, with an average year-to-year decrease in precipitation by 0.5-0.6%, and also in June, where the average year-to-year decrease in precipitation was estimated to be 0.9%. The results of the estimation are also presented visually in Fig. 3.15.

Regarding the residual diagnostics, one model (May) failed the test for no-autocorrelation in the *squared* residuals and the test for homoskedasticity. The homoskedasticity hypothesis was also rejected at the 5% significance level for two more months: April and August. It could indicate a potential violation of the IID assumption as the homoskedasticity hypothesis was violated for three out of twelve months, which could point at non-constant variance of the time series. However, the same test for the joint model for all 12 months in the following subsection did not reject the null hypothesis of homoskedasticity, nor were there other signs of misspecification. Therefore, it seems safe to rely on the results of the joint model discussed below.

	Jan	**Feb**	**Mar**	**Apr**	**May**	**Jun**
Intercept						
Coefficient	4.321	4.387	4.463	4.337	3.274	1.963
SE	0.070	0.067	0.055	0.061	0.106	0.093
t-test (p-val.)	0.000	0.000	0.000	0.000	0.000	0.000
Trend (t)						
Coefficient	-0.002	-0.006	-0.005	-0.006	-0.006	-0.005
SE	0.002	0.002	0.002	0.002	0.003	0.003
t-test (p-val.)	0.259	0.001	0.004	0.002	0.069	0.058
Residuals						
St. Dev.	0.264	0.253	0.207	0.231	0.404	0.352
Skewness	-0.726	-0.790	-0.105	-0.166	-0.608	-0.084
Kurtosis	3.959	3.957	2.889	2.788	2.749	2.944
Residuals – tests (p-values)						
Ljung-Box (5 lags):						
Residuals	0.698	0.184	0.750	0.123	0.487	0.154
Residuals^2	0.861	0.931	0.609	0.117	0.026	0.770
Turning point	0.678	0.917	0.917	0.917	0.917	0.407
Diff-sign	0.268	0.121	0.824	0.824	0.824	0.506
Goldfeld-Quandt	0.395	0.677	0.056	0.007	0.040	0.459
Jarque-Berra	0.023	0.014	0.932	0.824	0.146	0.962

	Jul	**Aug**	**Sep**	**Oct**	**Nov**	**Dec**
Intercept						
Coefficient	1.746	1.892	2.460	3.143	4.021	4.361
SE	0.090	0.082	0.082	0.094	0.099	0.090
t-test (p-val.)	0.000	0.000	0.000	0.000	0.000	0.000
Trend (t)						
Coefficient	-0.009	-0.000	-0.004	-0.001	-0.004	-0.003
SE	0.003	0.002	0.002	0.003	0.003	0.003
t-test (p-val.)	0.001	0.879	0.071	0.652	0.167	0.190
Residuals						
St. Dev.	0.340	0.309	0.310	0.355	0.376	0.340
Skewness	0.183	-0.265	-0.313	0.519	-0.263	-0.915
Kurtosis	2.116	2.461	3.074	2.825	2.458	3.830
Residuals – tests (p-values)						
Ljung-Box (5 lags):						
Residuals	0.493	0.009	0.411	0.754	0.140	0.076
Residuals^2	0.143	0.853	0.615	0.701	0.456	0.826
Turning point	0.147	0.017	0.678	0.678	0.097	0.097
Diff-sign	0.506	0.824	0.824	0.268	0.268	0.824
Goldfeld-Quandt	0.143	0.017	0.714	0.829	0.851	0.605
Jarque-Berra	0.318	0.489	0.609	0.251	0.489	0.006

Table 3.10: Tanzania, natural logarithm of monthly average precipitation. Results of trend analysis, OLS regression. Sample size for each regression was 60 years. See Fig. 3.15 below for plotted time series with estimated trends. (ERA5 data, Author's calculation).

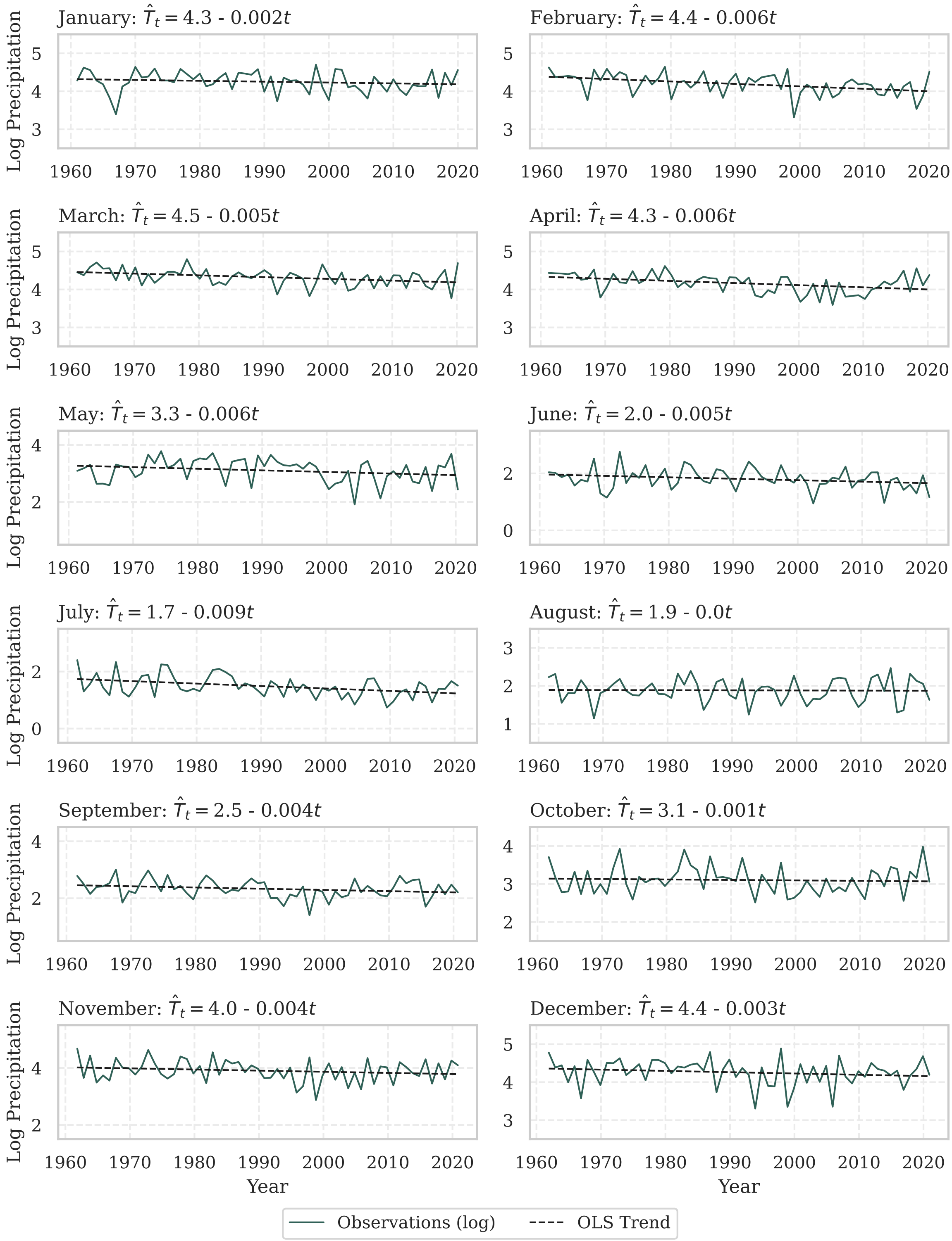

Figure 3.15: Natural logarithm of average monthly precipitation in Tanzania in 1961-2020 with estimated linear trends (OLS). See Tab. 3.10 above for detailed OLS estimation results (ERA5 data, Author's calculation).

Joint model for monthly precipitation

The results of the estimation of the joint model for the log of average monthly precipitation are reported in Tab. C.4; Fig. C.6 presents the results visually. The LR test did not reject the restriction that δ_i coefficients on all interaction terms in Eq. 3.5 are zero (p-value 0.24). Therefore, it seems that the trends in precipitation do not differ between the months of the year. The linear trend slope coefficient in the restricted model (Eq. 3.6) was estimated to be -0.004, and it was statistically significant. It indicates an average annual decrease in precipitation by 0.4%. There is also a clear seasonal pattern in precipitation, which is apparent in Fig. C.6. The restricted model does not seem to be misspecified: there seems to be no autocorrelation in the residuals nor the squared residuals up to the 12th lag (Ljung-Box p-values 0.875 and 0.367, respectively), and none of the other tests for the IID property rejects their null hypotheses. Residual plots are shown in Fig. C.7.

Summary of the chapter

The analysis in this chapter showed that the average temperatures in both countries are shifting upwards. This was apparent both from the descriptive part, where the hourly temperature distributions were investigated, and from the trend analysis. Warming is observed in all months of the year. The estimated average annual temperature change was larger for the Netherlands, but it is important to keep in mind that Tanzania is a hotter country, so seemingly lower shifts may prove even more detrimental for the country. There were no apparent changes in precipitation in the Netherlands, whereas, in Tanzania, a downward trend could be observed. Together with the higher temperatures, this could negatively impact agriculture and the livelihoods of people in general.

The limitations of the analysis presented in this chapter must be acknowledged. First of all, it used climate indicators aggregated over the entire country's territory. It could provide a high-level overview of the changes that have been happening, but a more detailed analysis on more granular data is required to build a more complete picture and to understand local challenges better. For example, it is worth investigating whether there are changes in the distribution of wet/dry and hot/cold periods, e.g. whether the precipitation accumulates in larger quantities to shorter periods, followed by extended dry periods. Furthermore, climate extremes are worth further investigation. Such analysis would require more granular local data since there are substantial regional differences in climatic indicators at any point in time within the countries.

Conclusion

In view of the pressing challenges for meeting food security targets, this thesis investigated the topics of agricultural production and productivity and changes in climatic indicators. The specific cases of two countries, the Netherlands and Tanzania, were scrutinised.

The first part of the thesis was focused on agricultural growth. In the Netherlands, which is known for its efficient agricultural sector, the largest changes occurred between 1961 and 1980, with a significant role of substituting labour with capital and a positive TFP growth rate. In Tanzania, a Sub-Saharan country where agriculture is still vital for the livelihoods of households, agricultural output growth was mostly resource-driven. In the last three decades, growth in capital and materials employed in agriculture contributed substantially to expanding agricultural production. At the same time, the estimated TFP growth rate was negative, indicating a deteriorating efficiency of the inputs employed in agriculture. Lastly, the vast gap in agricultural efficiency and capacities between these countries was noted.

The second part of the analysis focused on the changes in average temperatures and precipitation. Upward shifts in average temperatures were detected in both countries, while a decreasing trend in precipitation was also found in Tanzania. The temperatures were estimated to grow annually by 0.032°C in the Netherlands and 0.018°C in Tanzania. Precipitation in Tanzania was found to decrease, on average, by 0.4% per year. A potential issue of volatility clustering was detected in the monthly temperature time series and is worth further investigation as it violates the IID assumption imposed on the error term by the models. Nonetheless, it is apparent from the analysis that climatic changes can be observed in both countries. However, the extent of the consequences may be different. It would require a more detailed investigation on a more granular level than the level of a country.

It is evident that farmers in these two countries operate under very different circumstances: large-scale mechanised agriculture, with an easily accessible European market and a long culture of entrepreneurship and innovativeness with support from the policy on the one hand, versus developing economy where the green revolution is yet expected to occur, agriculture is dominated by smallholder farmers with resource and information constraints, weak institutional support and lack of access to the market, on the other hand. Under such different circumstances, the difference in the degree of vulnerability to the upcoming threats of climate change and the capacity to adapt is clear. In fact, African farmers already seem to be feeling the pressure from the changing climate: 34% of the Tanzanian farmer households mentioned climate change (droughts, floods etc.) among their top-3 constraints for agricultural production in the latest agricultural census, climate change being second only to the cost of inputs (66).

The analysis of agricultural and climatic variables in this work laid the basis for further research on the relationships between agriculture and climate change. The agricultural part of the analysis shed light on the channels through which changes in climate could affect agriculture. For example, increased heat may negatively affect labour productivity (77), which may be especially pronounced for agricultural workers who have to spend much time outside in the fields. Land as an agricultural resource could be affected if the locations with climatic conditions suitable for agriculture were to shift, as can be expected under certain scenarios (93). In some areas, the growing season may have become longer, potentially positively impacting agricultural output. Potential impacts on modern inputs such as fertilisers are also worth investigating. Furthermore, more frequent climate extremes could abruptly affect every one of these dimensions at once. Therefore, it is imperative that the most vulnerable spots are identified and targeted by suitable interventions. For these purposes, the risks at a local level need to be investigated.

The issue is indisputably complex, and these effects may be hard to disentangle in the data. Furthermore, causal inference is a complicated task, and it can be argued that it is impossible to derive true causal relationships from observational data. With climate change, humanity has not observed a counterfactual scenario of no warming, so it is impossible to quantify its true effects on agriculture and other dimensions. Aside from climatic factors, other factors, such as institutions, play an important role in the course of development. However, given the pressure of the growing food demand and the risks to food security, every effort should be taken to improve our understanding of these complex relationships. These questions remain open for further research.

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Appendices

A. Supplementary figures and tables to Chapter 2

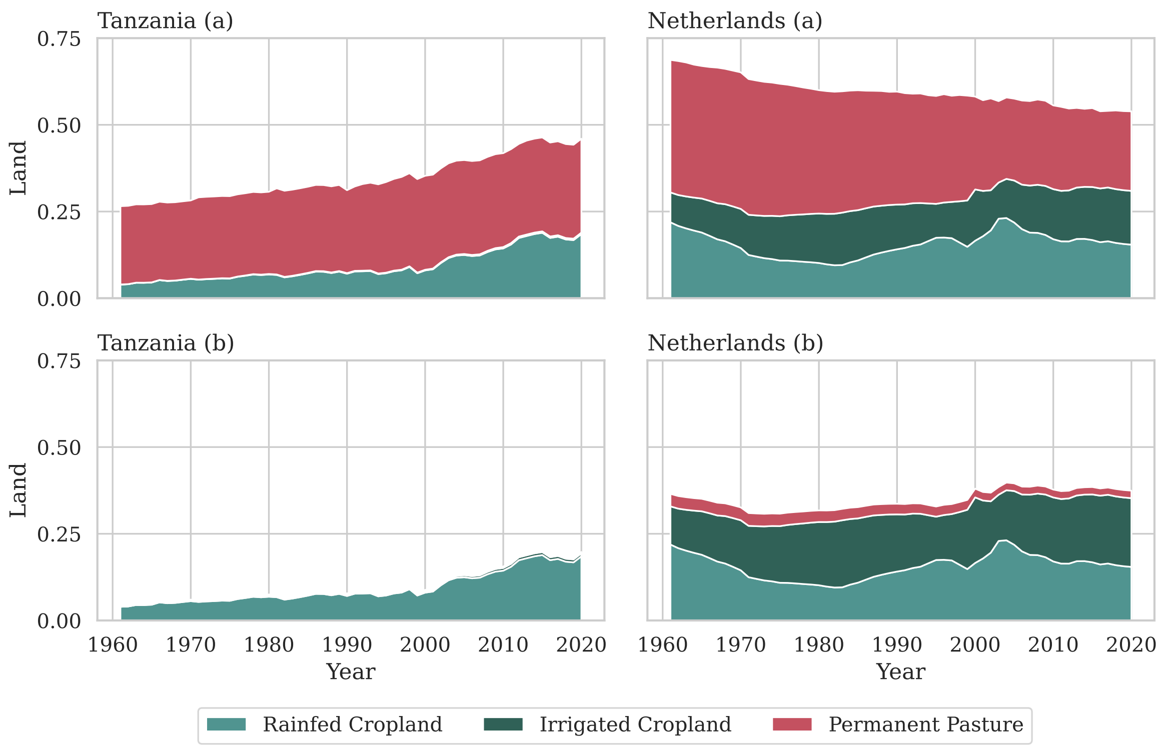

Figure A.1: Agricultural land by type: rainfed cropland, irrigated cropland and permanent pastures, standardised by the total land area of the country; (a) total area, (b) quality-adjusted land as it enters inputs aggregation (Agricultural data from USDA, country territory from FAO, Author's calculation).

	Unit	1961	1970	1980	1990	2000	2010	2020
Agricultural inputs								
Cropland	[1000 ha]	1027	867	822	909	1056	1059	1042
Irrig. cropland	[% of total]	28.2	43.8	58.4	47.7	47.2	45.9	50.1
Pastures	[1000 ha]	1287	1326	1198	1097	902	813	772
Labour	[1000 pers.]	476.0	314.3	244.0	248.2	241.7	229.4	166.3
Capital per w.	[$ 2015]	36.59	105.31	180.21	178.19	180.86	223.41	364.90
Machinery per w.	[hp]	5.45	17.97	29.91	29.12	25.45	24.63	29.71
Fertilizer per ha	[kg]	652.6	1009.9	1221.7	988.4	685.3	568.8	514.3
Agricultural output								
Output total	[B$ 2015]	8.060	10.827	14.570	17.085	17.165	16.334	18.518
Per capita	[$ 2015]	692.0	830.4	1031.1	1143.2	1079.6	983.0	1062.2
Output structure								
Crop	[% of total]	31.5	29.8	23.7	25.9	29.0	30.7	28.7
Animal	[% of total]	66.7	68.8	75.4	73.1	70.2	68.6	70.9
Fishery	[% of total]	1.8	1.3	0.9	1.0	0.8	0.8	0.4
Agricultural productivity								
Crop yield	[K$ 2015 per ha]	2.5	3.7	4.2	4.9	4.7	4.7	5.1
Output per worker	[K$ 2015 per w.]	16.9	34.5	59.7	68.9	71.0	71.2	111.4

Table A.1: The Netherlands. Summary of the main agricultural indicators in levels. Agricultural data from USDA, population data for per capita calculation from UN WPP, Author's calculation. *Notes:* per w. = per worker; $ 2015 = 2015 international dollars; B = billions, K = thousands. Crop yield is calculated as crop output divided by the cropland area; output per worker is total agricultural output divided by the number of workers in agriculture. Land area in raw values, not quality-adjusted.

	1961-1970	**1971-1980**	**1981-1990**	**1991-2000**	**2001-2010**	**2011-2020**
Agricultural inputs						
Aggregate Inputs	**0.88**	**1.67**	**0.41**	**-0.39**	**1.03**	**0.57**
Land (q-adj.)	**-1.18**	**0.33**	**0.74**	**0.86**	**0.27**	**0.01**
Rainfed cropland	-4.43	-2.13	4.91	0.85	-1.58	-0.83
Irrig. cropland	3.00	2.30	-1.63	1.19	2.62	0.78
Pastures	0.29	-1.08	-1.03	-1.30	-0.53	-0.53
Labour	**-4.88**	**-2.54**	**-0.51**	**-0.90**	**0.43**	**-1.86**
Capital	**7.08**	**2.72**	**0.10**	**-0.16**	**1.60**	**0.87**
Capital per w.	11.96	5.26	0.61	0.73	1.16	2.73
Machinery	8.53	2.38	-0.02	-1.70	-0.83	-1.13
Machinery per w.	13.41	4.92	0.49	-0.80	-1.26	0.73
Livestock	1.62	4.02	0.61	-1.28	0.25	-0.45
Materials	**1.82**	**2.96**	**0.63**	**-0.54**	**1.10**	**1.08**
Fertilizer	2.45	2.00	-1.04	-2.00	-1.78	-0.52
Fertilizer per ha	4.23	1.74	-2.37	-3.02	-1.85	-0.46
Animal feed	1.25	3.84	2.78	0.65	3.15	2.21
Agricultural output						
Total	**2.98**	**3.17**	**1.11**	**0.25**	**0.40**	**1.12**
Crop	2.65	0.92	2.44	0.56	1.03	0.29
Animal	3.20	4.08	0.73	0.04	0.15	1.48
Fishery	0.49	-1.83	-1.59	7.02	-0.88	-0.35
Agricultural productivity						
TFP	**2.10**	**1.50**	**0.70**	**0.64**	**-0.63**	**0.55**
Crop yield	4.43	0.68	1.11	-0.46	0.96	0.35
Output per worker	7.86	5.70	1.62	1.15	-0.03	2.98

Table A.2: The Netherlands. Annual growth rates of the main agricultural indicators by decade, in % per year. Agricultural data from USDA, Author's calculation. *Notes:* per w. = per worker. Total land growth rate is reported for the quality-adjusted aggregated land, growth rates for the land types were calculated on the raw values.

	Unit	1961	1970	1980	1990	2000	2010	2020
Agricultural inputs								
Cropland	[1000 ha]	3513	4976	6188	6406	7269	13058	16746
Irrig. cropland	[% of total]	0.6	0.8	1.9	2.3	2.2	2.6	2.2
Pastures	[1000 ha]	20000	20000	21000	21190	24000	24000	24000
Labour	[1000 pers.]	4571	5725	7209	9694	12881	14380	17694
Capital per w.	[$ 2015]	0.17	0.16	0.15	0.12	0.14	0.24	0.53
Machinery per w.	[hp]	0.15	0.12	0.06	0.03	0.05	0.07	0.07
Fertilizer per ha	[kg]	5.5	7.3	9.8	12.3	8.5	14.2	17.9
Agricultural output								
Output total	[B$ 2015]	2.4	3.3	4.3	5.7	6.5	11.5	17.6
Per capita	[$ 2015]	229.6	238.7	222.0	218.8	187.9	256.0	284.8
Output structure								
Crop	[% of total]	72.8	73.4	76.8	74.2	70.1	78.0	75.0
Animal	[% of total]	27.2	26.6	23.2	25.8	30.0	21.9	24.7
Fishery	[% of total]	0.0	0.0	0.0	0.0002	0.0001	0.0004	0.0022
Agricultural productivity								
Crop yield	[K$ 2015 per ha]	0.49	0.48	0.53	0.66	0.62	0.69	0.79
Output per worker	[K$ 2015 per w.]	0.52	0.57	0.59	0.59	0.50	0.80	0.99

Table A.3: Tanzania. Summary of the main agricultural indicators in levels. Agricultural data from USDA, population data for per capita calculation from UN WPP, Author's calculation. *Notes:* per w. = per worker; $ 2015 = 2015 international dollars; B = billions, K = thousands. Crop yield is calculated as crop output divided by the cropland area; output per worker is total agricultural output divided by the number of workers in agriculture. Land area in raw values, not quality-adjusted.

	1961-1970	1971-1980	1981-1990	1991-2000	2001-2010	2011-2020
Agricultural inputs						
Aggregate inputs	**3.02**	**2.87**	**1.72**	**2.37**	**5.08**	**4.74**
Land (q-adj.)	**3.52**	**3.15**	**1.90**	**0.66**	**4.74**	**0.49**
Rainfed cropland	3.74	3.08	2.00	0.62	4.78	0.51
Irrig. cropland	7.07	12.94	2.15	0.83	7.67	0.21
Pastures	0.00	0.00	-0.20	1.17	0.00	0.06
Labour	**2.50**	**2.30**	**2.95**	**2.83**	**0.95**	**2.26**
Capital	**2.21**	**1.67**	**0.06**	**5.35**	**6.34**	**10.41**
Capital per w.	-0.28	-0.63	-2.89	2.51	5.40	8.15
Machinery	0.34	-5.76	-3.13	9.01	2.03	2.12
Machinery per w.	-2.15	-8.06	-6.08	6.17	1.08	-0.14
Livestock	2.36	2.25	0.31	2.41	1.68	2.49
Materials	**5.34**	**7.22**	**3.49**	**-3.07**	**11.92**	**5.44**
Fertilizer	7.34	4.94	4.09	-2.74	8.57	5.02
Fertilizer per ha	3.58	1.73	2.08	-3.37	3.72	4.51
Animal feed	3.10	9.79	2.81	-3.45	15.70	5.92
Agricultural output						
Total	**3.29**	**4.09**	**2.61**	**1.37**	**4.76**	**3.14**
Crop	3.25	4.65	2.21	1.21	5.10	2.50
Animal	3.43	2.41	3.82	1.77	3.73	5.20
Fishery	0.00	0.00	186.48	-2.59	15.43	21.05
Agricultural productivity						
TFP	**0.27**	**1.22**	**0.89**	**-1.00**	**-0.32**	**-1.60**
Crop yield	4.43	0.68	1.11	-0.46	0.96	0.35
Output per worker	7.86	5.70	1.62	1.15	-0.03	2.98

Table A.4: Tanzania. Annual growth rates of the main agricultural indicators by decade, in % per year. Agricultural data from USDA, Author's calculation. *Notes:* per w. = per worker. Total land growth rate is reported for the quality-adjusted aggregated land, growth rates for the land types were calculated on the raw values.

B. Python code for climatic data retrieval from CDS Toolbox

```python
"""
This script is intended for online CDS Toolbox editor:
https://cds.climate.copernicus.eu/toolbox/doc/index.html

It retrieves data on total precipitation from ERA5-Land reanalysis
and aggregates it over the territory of specified country
(the Netherlands / Tanzania)
"""
# Notes:
# (1)
# requests take a long time, so it is better to
# retrieve data in chunks (e.g. by 20 years)
# (2)
# CDS toolbox allows to retrieve 1 month at a time,
# that is why there is an additional loop in the code


import cdstoolbox as ct

# area for data retrieval for countries
areas = {
    'TZA': [0, 29, -12, 41], # Tanzania
    'NLD': [0, 29, -12, 41]  # Netherlands
}
# specify ISO3 of the country:
country = 'TZA' # ['TZA', 'NLD']

# specify the variable
varnm = 'total_precipitation'
"""
variables = {
    'Near-Surface Air Temperature': '2m_temperature',
    'Total Precipitation': 'total_precipitation',
}
"""

@ct.application()
@ct.output.download(format='csv')
def compute_avg():
    data_list = []
    # define year range
    years = list(range(1961, 1980 + 1))
    months = list(range(1, 12 + 1))

    # retrieve and transform data in one-month chunks
    for year in years:
        for month in months:
```

```
            # 1. Retrieve raw data
            data = ct.catalogue.retrieve(
                'reanalysis-era5-land',
                {
                    'variable': varnm,
                    'year': year,
                    'month': month,
                    'day': list(range(1, 31 + 1)),
                    'time': [
                        '00:00', '01:00', '02:00', '03:00',
                        '04:00', '05:00', '06:00', '07:00',
                        '08:00', '09:00', '10:00', '11:00',
                        '12:00', '13:00', '14:00', '15:00',
                        '16:00', '17:00', '18:00', '19:00',
                        '20:00', '21:00', '22:00', '23:00'
                    ],
                    # limit area for faster retrieval
                    'area': areas[country]
                }
            )
            # 2. Select only grid cells on the country territory
            country_shape = ct.shapes.catalogue.countries(ISO_A3=country)
            data_masked = ct.shapes.mask(data, country_shape, drop=True)
            # 3. Calculate spatial average over the country territory
            country_avg = ct.geo.spatial_average(data_masked)
            # 4. For temperature - convert to degrees Celsius;
            #    Store transformed data array in a list
            if varnm=='2m_temperature':
                res_cels = ct.cdm.convert_units(country_avg, 'Celsius')
                data_list.append(res_cels)
            else:
                data_list.append(country_avg)

    # Concatenate all chunks into one array and return result
    result = ct.cube.concat(data_list, dim='time')
    return result
```

C. Supplementary figures and tables to Chapter 3

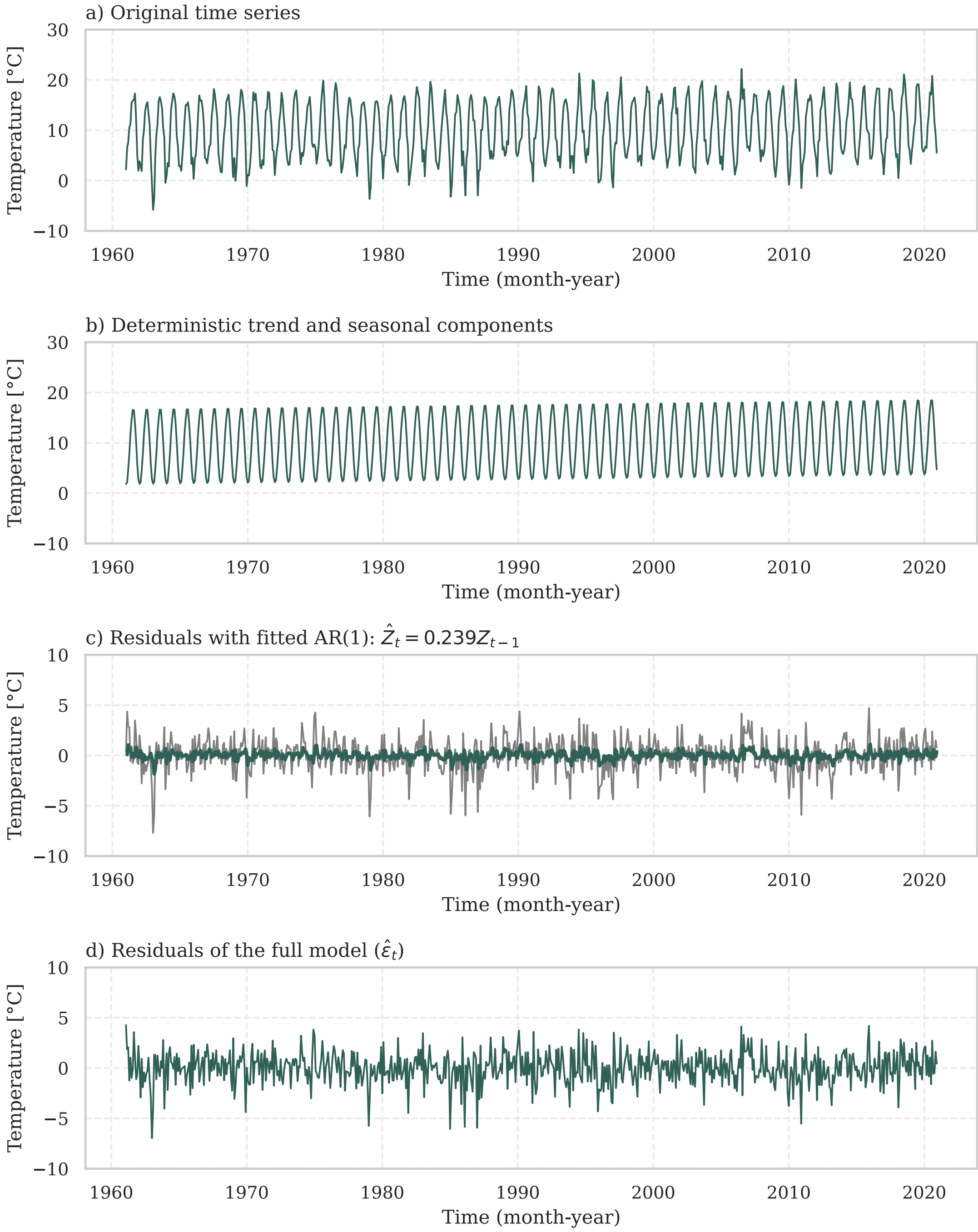

Figure C.1: Average monthly temperatures in the Netherlands in 1961-2020 with estimated deterministic trend and seasonal components and $AR(1)$ residual. Visualisation of estimation results, GLS restricted model, see Tab. C.1 for detailed results and Fig. C.2 for residual plots (ERA5 data, Author's calculation).

	OLS (U)	**OLS (R)**	**GLS (U)**	**GLS(R)**
Intercept	2.499^{***}	2.782^{***}	2.508^{***}	2.783^{***}
Year (t_t^*)	0.042^{***}	0.032^{***}	0.041^{***}	0.032^{***}
Seasonal dummies				
Jan	-0.930	-1.003^{***}	-0.944^{*}	-1.001^{***}
Feb	-0.305	-0.638^{*}	-0.316	-0.639^{**}
Mar	2.108^{***}	1.803^{***}	2.098^{***}	1.802^{***}
Apr	4.943^{***}	4.970^{***}	4.933^{***}	4.969^{***}
May	9.292^{***}	8.999^{***}	9.283^{***}	8.998^{***}
Jun	12.297^{***}	11.872^{***}	12.287^{***}	11.871^{***}
Jul	13.729^{***}	13.732^{***}	13.719^{***}	13.731^{***}
Aug	13.809^{***}	13.685^{***}	13.799^{***}	13.683^{***}
Sep	11.740^{***}	11.003^{***}	11.731^{***}	11.002^{***}
Oct	8.200^{***}	7.383^{***}	8.191^{***}	7.381^{***}
Nov	3.258^{***}	2.928^{***}	3.251^{***}	2.926^{***}
Interactions				
Jan*Year	-0.002		-0.002	
Feb*Year	-0.011		-0.011	
Mar*Year	-0.010		-0.010	
Apr*Year	0.001		0.001	
May*Year	-0.010		-0.009	
Jun*Year	-0.014		-0.014	
Jul*Year	0.000		0.000	
Aug*Year	-0.004		-0.004	
Sep*Year	-0.024		-0.024	
Oct*Year	-0.027		-0.027	
Nov*Year	-0.011		-0.011	
Joint significance (LR test)				
Log Likelihood			-1354.2	-1357.2
LR p-value			0.874	
AR process for residual Z_t				
Z_{t-1}			0.238^{***}	0.239^{***}
Residuals ($\hat{Z}_t$ for OLS, $\hat{\epsilon}_t$ for GLS)				
St. Dev.	1.635	1.642	1.589	1.596
Skew	-0.488	-0.487	-0.407	-0.411
Kurt	4.471	4.471	4.199	4.196
Residuals – tests (p-values)				
Ljung-Box (12 lags)				
Residuals	0.000	0.000	0.856	0.883
Residuals^2	0.000	0.000	0.000	0.000
Turning point	0.157	0.021	0.595	0.859
Diff-sign	0.897	0.438	0.366	0.439
Goldfeld–Quandt	0.506	0.434	0.808	0.719
Jarque-Berra	0.000	0.000	0.000	0.000

Table C.1: The Netherlands, monthly average temperature. Results of joint trend and seasonal model estimation. GLS estimation of unrestricted (U) model (Eq. 3.5) and restricted (R) model (Eq. 3.6). OLS regression output is given for comparison (HAC standard errors used for inference). Sample size for estimation was 720 periods (60 years * 12 months). Significance levels: * 10%, ** 5%, *** 1% (ERA5 data, Author's calculation).

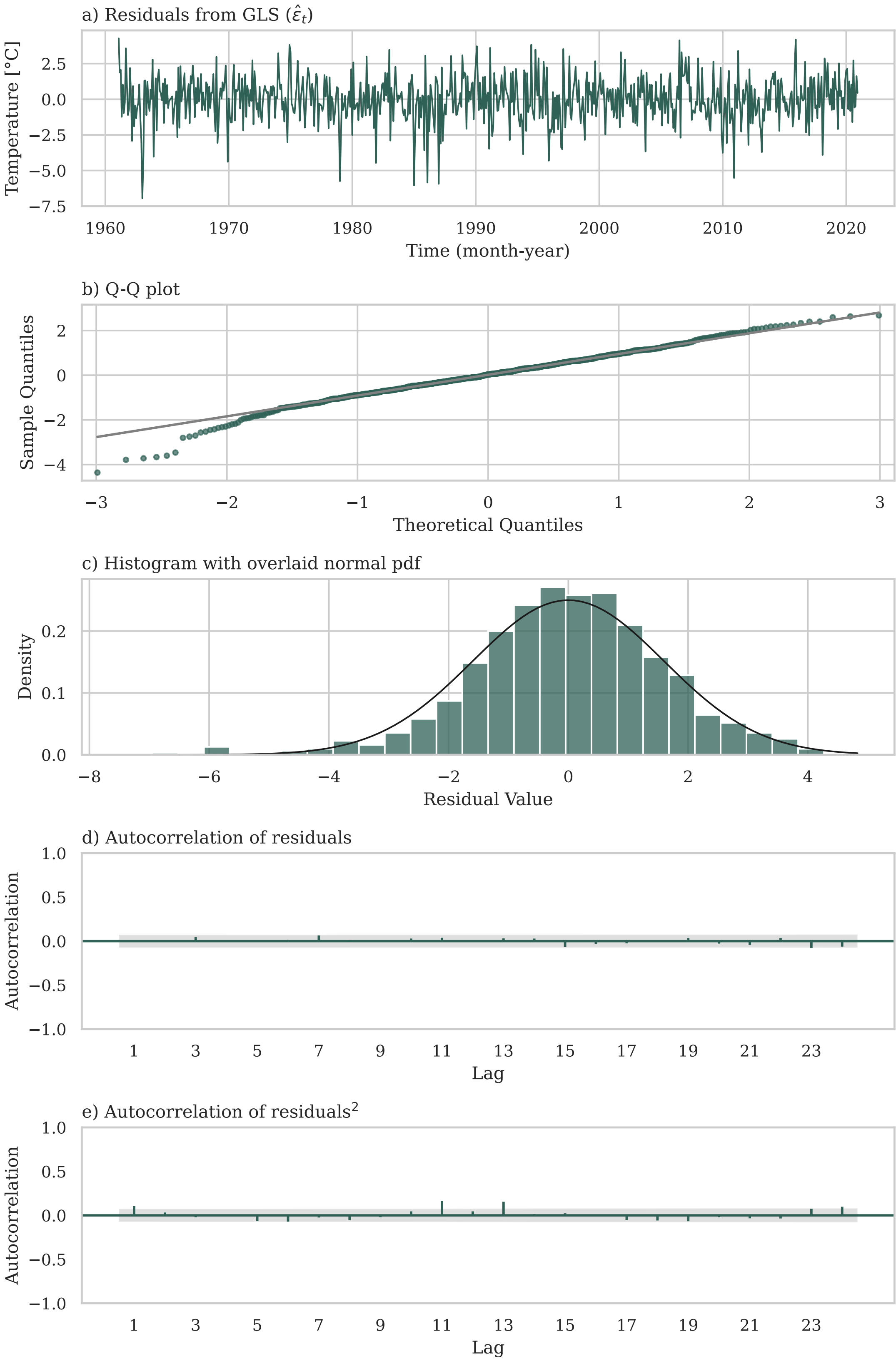

Figure C.2: Average monthly temperatures in the Netherlands in 1961-2020. Residual plots for GLS restricted model with deterministic trend and seasonal components and $AR(1)$ residual presented in Fig. C.1 and Tab. C.1 above (ERA5 data, Author's calculation).

Month	**Deviation**	**Mean**	**St.Dev.**	**Median**
January	-0.75	27.78	11.62	27.42
February	-4.14	24.39	12.20	23.83
March	-2.14	26.39	11.32	26.78
April	-5.03	23.49	10.97	21.88
May	-0.86	27.67	10.32	26.55
June	2.05	30.57	12.06	29.64
July	4.53	33.05	14.41	31.75
August	3.07	31.59	13.02	29.98
September	0.08	28.61	13.01	25.51
October	-0.47	28.05	12.74	26.34
November	1.33	29.85	10.74	30.81
December	2.32	30.85	13.43	29.09
All		28.52	12.42	27.38

Table C.2: The Netherlands, monthly time series of average daily precipitation (in levels, not logs). Seasonal deviations and descriptive statistics. Seasonal deviations from the mean were derived from OLS estimation, where monthly average precipitation was regressed on a constant and eleven seasonal dummies. See Fig. C.3 for visualisation (ERA5 data, Author's calculation).

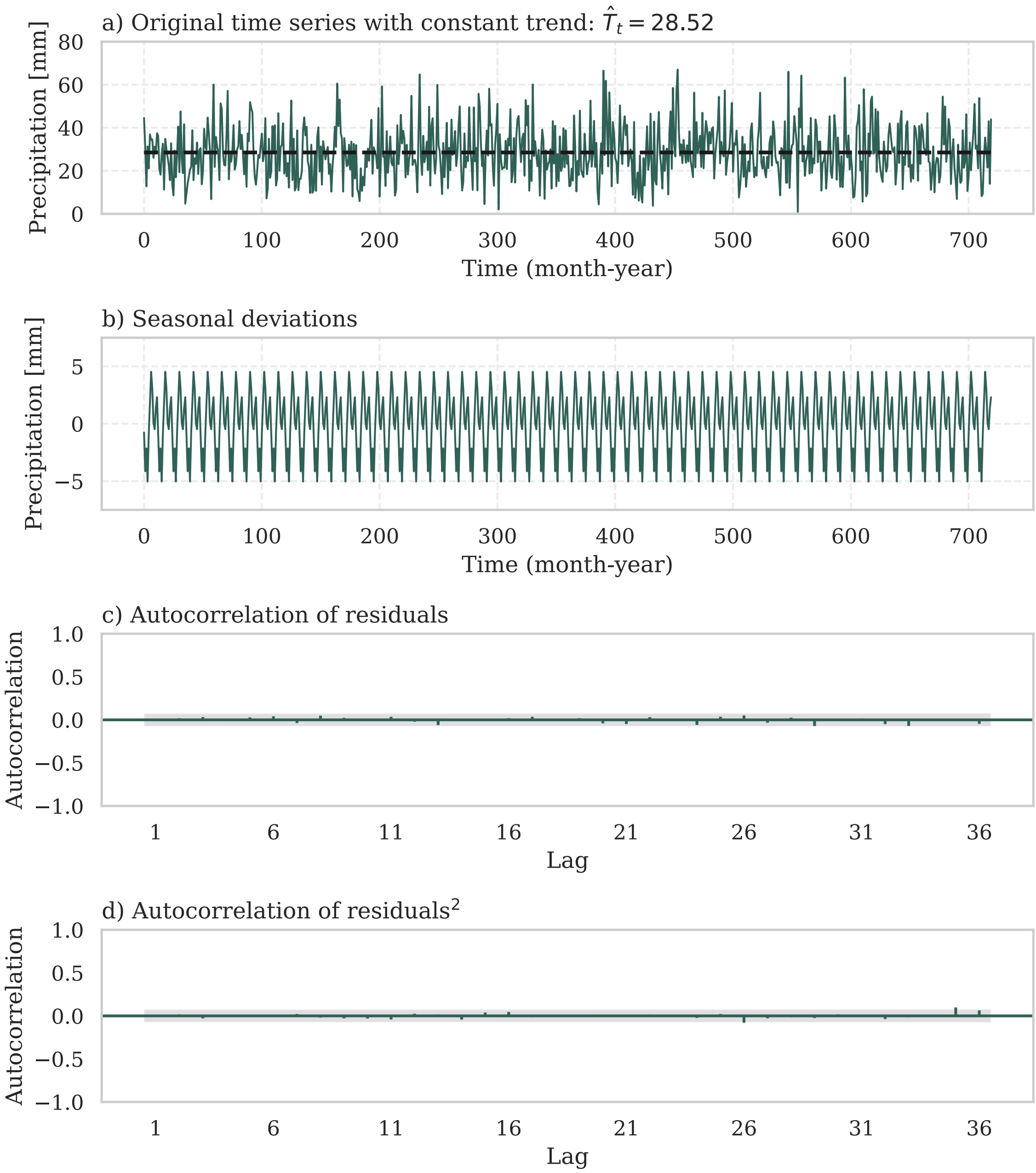

Figure C.3: The Netherlands, monthly time series of average daily precipitation (in levels, not logs). Time series with estimated seasonal deviations and constant trend, OLS estimation. See Tab. C.2 for precise values of seasonal deviations (ERA5 data, Author's calculation).

	OLS (U)	**OLS (R)**	**GLS (U)**	**GLS(R)**
Intercept	21.918^{***}	22.095^{***}	21.954^{***}	22.110^{***}
Year (t_t^*)	0.024^{***}	0.019^{***}	0.023^{***}	0.018^{***}
Seasonal dummies				
Jan	-0.079	-0.128	-0.108	-0.136^{**}
Feb	0.011	0.100	-0.020	0.091
Mar	-0.117	-0.152	-0.149	-0.159^{**}
Apr	-0.553^{***}	-0.767^{***}	-0.585^{***}	-0.774^{***}
May	-0.824^{***}	-1.000^{***}	-0.856^{***}	-1.008^{***}
Jun	-1.280^{***}	-1.606^{***}	-1.312^{***}	-1.613^{***}
Jul	-1.459^{***}	-1.924^{***}	-1.490^{***}	-1.931^{***}
Aug	-0.766^{***}	-1.104^{***}	-0.794^{***}	-1.110^{***}
Sep	0.415^{**}	0.131	0.390^{**}	0.125
Oct	1.175^{***}	0.939^{***}	1.156^{***}	0.935^{***}
Nov	0.745^{***}	0.658^{***}	0.736^{***}	0.657^{***}
Interactions				
Jan*Year	-0.002		-0.001	
Feb*Year	0.003		0.004	
Mar*Year	-0.001		0.000	
Apr*Year	-0.007		-0.006	
May*Year	-0.006		-0.005	
Jun*Year	-0.011^{*}		-0.010^{**}	
Jul*Year	-0.015^{***}		-0.014^{***}	
Aug*Year	-0.011^{**}		-0.010^{**}	
Sep*Year	-0.009^{*}		-0.009^{*}	
Oct*Year	-0.008		-0.007^{*}	
Nov*Year	-0.003		-0.003	
Joint significance (LR test)				
Log Likelihood			-376.4	-385.4
LR p-value			0.082	
AR process for residual Z_t				
Z_{t-1}			0.444^{***}	0.454^{***}
Z_{t-2}			0.127^{***}	0.119^{***}
Residuals ($\hat{Z}_t$ ***for OLS,*** $\hat{\epsilon}_t$ ***for GLS)***				
St. Dev.	0.477	0.486	0.405	0.410
Skew	0.485	0.490	0.161	0.153
Kurt	3.705	3.640	3.964	3.980
Residuals – tests (p-values)				
Ljung-Box (12 lags)				
Residuals	0.000	0.000	0.025	0.190
Residuals^2	0.000	0.000	0.000	0.001
Turning point	0.000	0.000	0.204	0.460
Diff-sign	0.004	0.045	0.106	0.061
Goldfeld–Quandt	0.263	0.082	0.551	0.299
Jarque-Berra	0.000	0.000	0.000	0.000

Table C.3: Tanzania, monthly average temperature. Results of joint trend and seasonal model estimation. GLS estimation of unrestricted (U) model (Eq. 3.5) and restricted (R) model (Eq. 3.6). OLS regression output is given for comparison (HAC standard errors used for inference). Sample size for estimation was 720 periods (60 years * 12 months). Significance levels: * 10%, ** 5%, *** 1% (ERA5 data, Author's calculation).

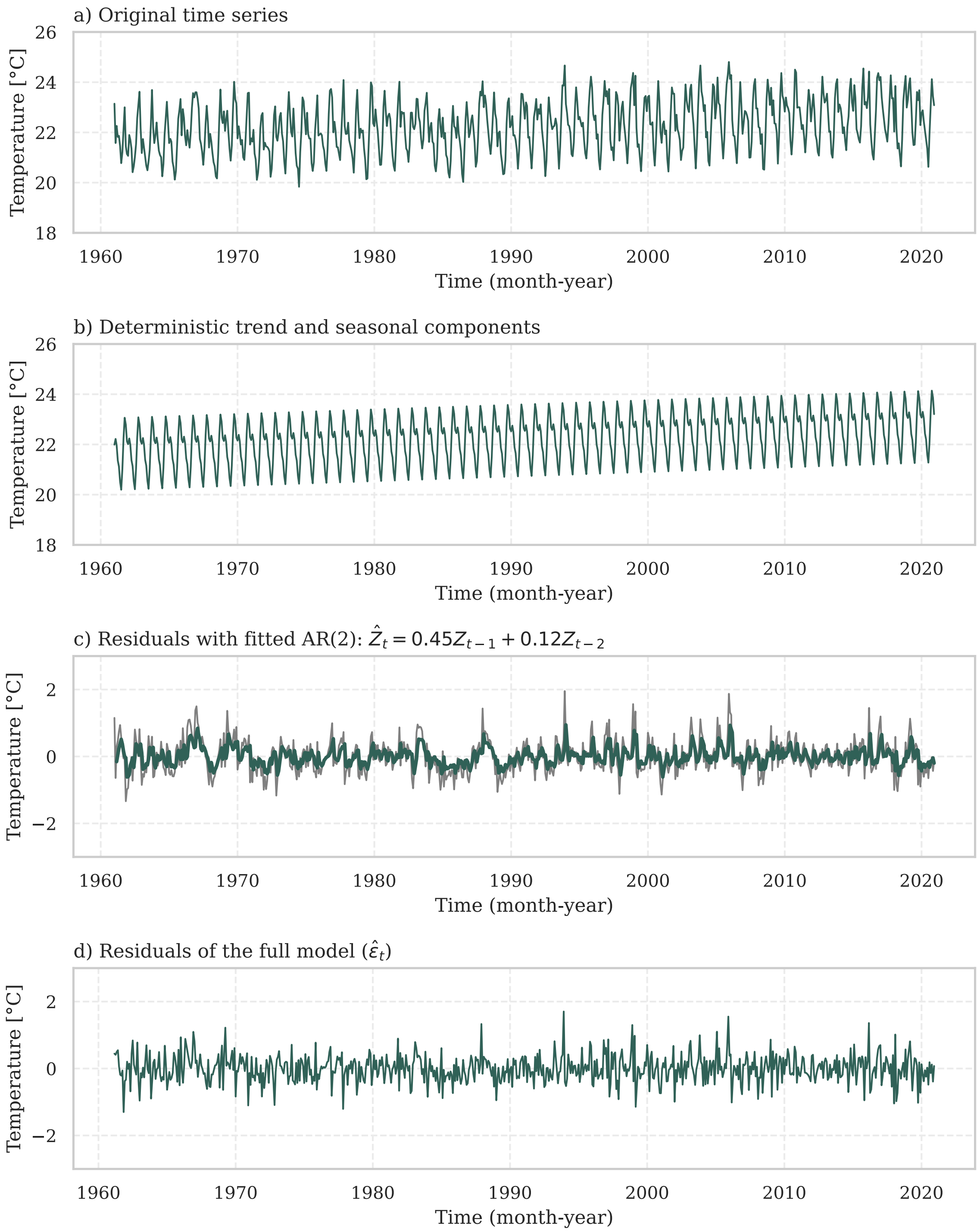

Figure C.4: Average monthly temperatures in Tanzania in 1961-2020 with estimated deterministic trend and seasonal components and $AR(2)$ residual. Visualisation of estimation results, GLS restricted model, see Tab. C.3 for detailed results and Fig. C.5 for residual plots (ERA5 data, Author's calculation).

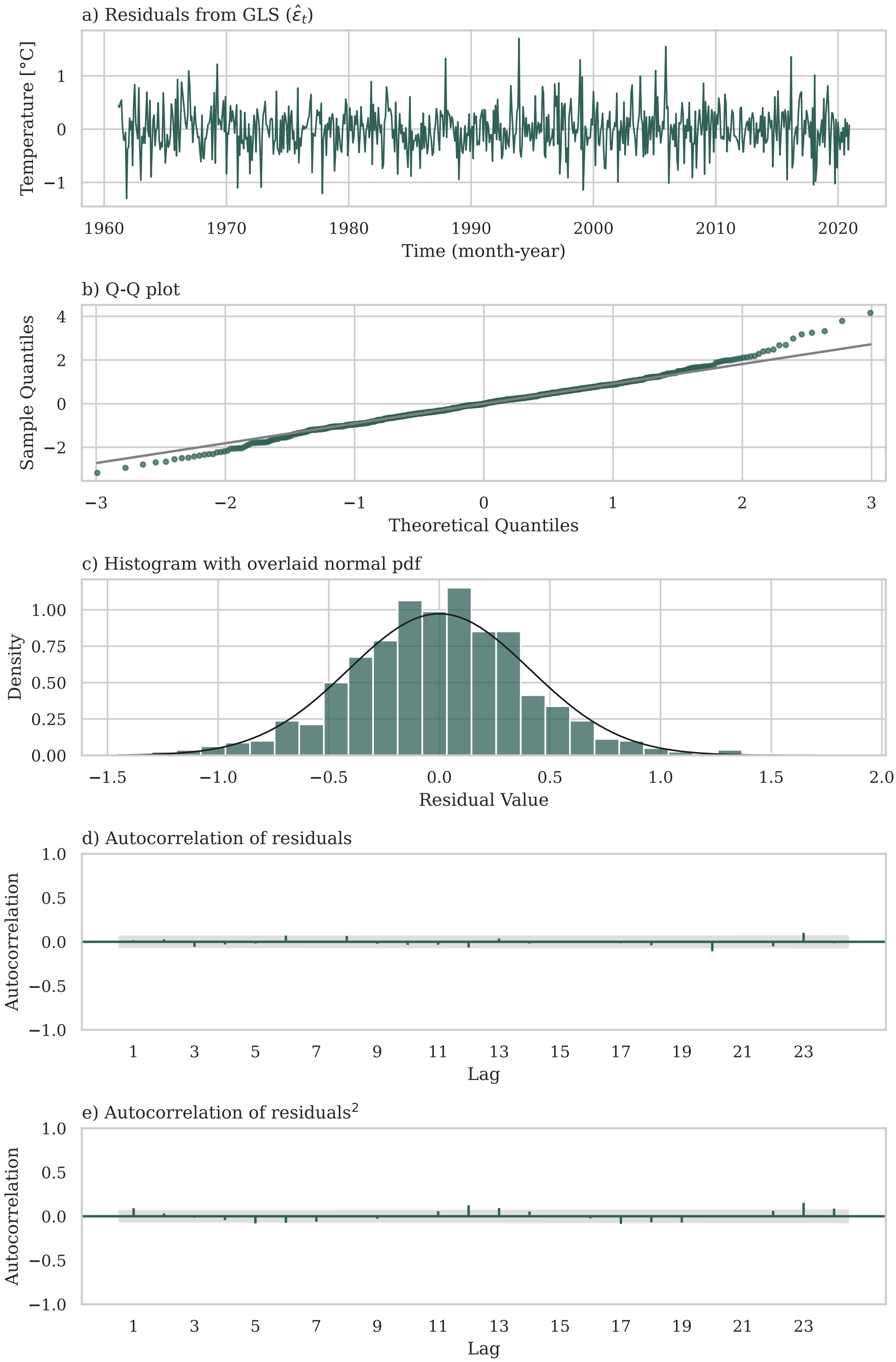

Figure C.5: Average monthly temperatures in Tanzania in 1961-2020. Residual plots for GLS restricted model with deterministic trend and seasonal components and $AR(2)$ residual presented in Fig. C.4 above (ERA5 data, Author's calculation).

	OLS (U)	**OLS (R)**	**GLS (U)**	**GLS(R)**
Intercept	4.361^{***}	4.389^{***}	4.362^{***}	4.388^{***}
Year (t_t^*)	-0.003	-0.004^{***}	-0.003	-0.004^{***}
Seasonal dummies				
Jan	-0.040	-0.006	-0.043	-0.005
Feb	0.026	-0.067	0.024	-0.067
Mar	0.102	0.065	0.101	0.065
Apr	-0.025	-0.093^{*}	-0.026	-0.093
May	-1.088^{***}	-1.156^{***}	-1.089^{***}	-1.156^{***}
Jun	-2.398^{***}	-2.450^{***}	-2.399^{***}	-2.450^{***}
Jul	-2.616^{***}	-2.774^{***}	-2.616^{***}	-2.774^{***}
Aug	-2.469^{***}	-2.377^{***}	-2.470^{***}	-2.377^{***}
Sep	-1.901^{***}	-1.928^{***}	-1.902^{***}	-1.928^{***}
Oct	-1.219^{***}	-1.152^{***}	-1.220^{***}	-1.152^{***}
Nov	-0.341^{***}	-0.358^{***}	-0.341^{***}	-0.358^{***}
Interactions				
Jan*Year	0.001		0.001	
Feb*Year	-0.003		-0.003	
Mar*Year	-0.001		-0.001	
Apr*Year	-0.002		-0.002	
May*Year	-0.002		-0.002	
Jun*Year	-0.002		-0.002	
Jul*Year	-0.005		-0.005	
Aug*Year	0.003		0.003	
Sep*Year	-0.001		-0.001	
Oct*Year	0.002		0.002	
Nov*Year	-0.001		-0.001	
Joint significance (LR test)				
Log Likelihood			-135.2	-142.1
LR p-value			0.241	
AR process for residual Z_t				
Z_{t-1}			0.326^{***}	0.316^{***}
Z_{t-2}			0.096^{***}	0.104^{***}
Residuals ($\hat{Z}_t$ for OLS, $\hat{\epsilon}_t$ for GLS)				
St. Dev.	0.315	0.317	0.292	0.295
Skew	-0.286	-0.257	-0.249	-0.222
Kurt	3.222	3.310	3.198	3.293
Residuals – tests (p-values)				
Ljung-Box (12 lags)				
Residuals	0.000	0.000	0.653	0.875
Residuals^2	0.001	0.006	0.214	0.367
Turning point	0.001	0.004	0.274	0.274
Diff-sign	0.220	0.651	0.061	0.175
Goldfeld-Quandt	0.444	0.390	0.292	0.329
Jarque-Berra	0.004	0.005	0.014	0.014

Table C.4: Tanzania, log of monthly average precipitation. Results of joint trend and seasonal model estimation. GLS estimation of unrestricted (U) model (Eq. 3.5) and restricted (R) model (Eq. 3.6). OLS regression output is given for comparison (HAC standard errors used for inference). Sample size for estimation was 720 periods (60 years * 12 months). Significance levels: * 10%, ** 5%, *** 1% (ERA5 data, Author's calculation).

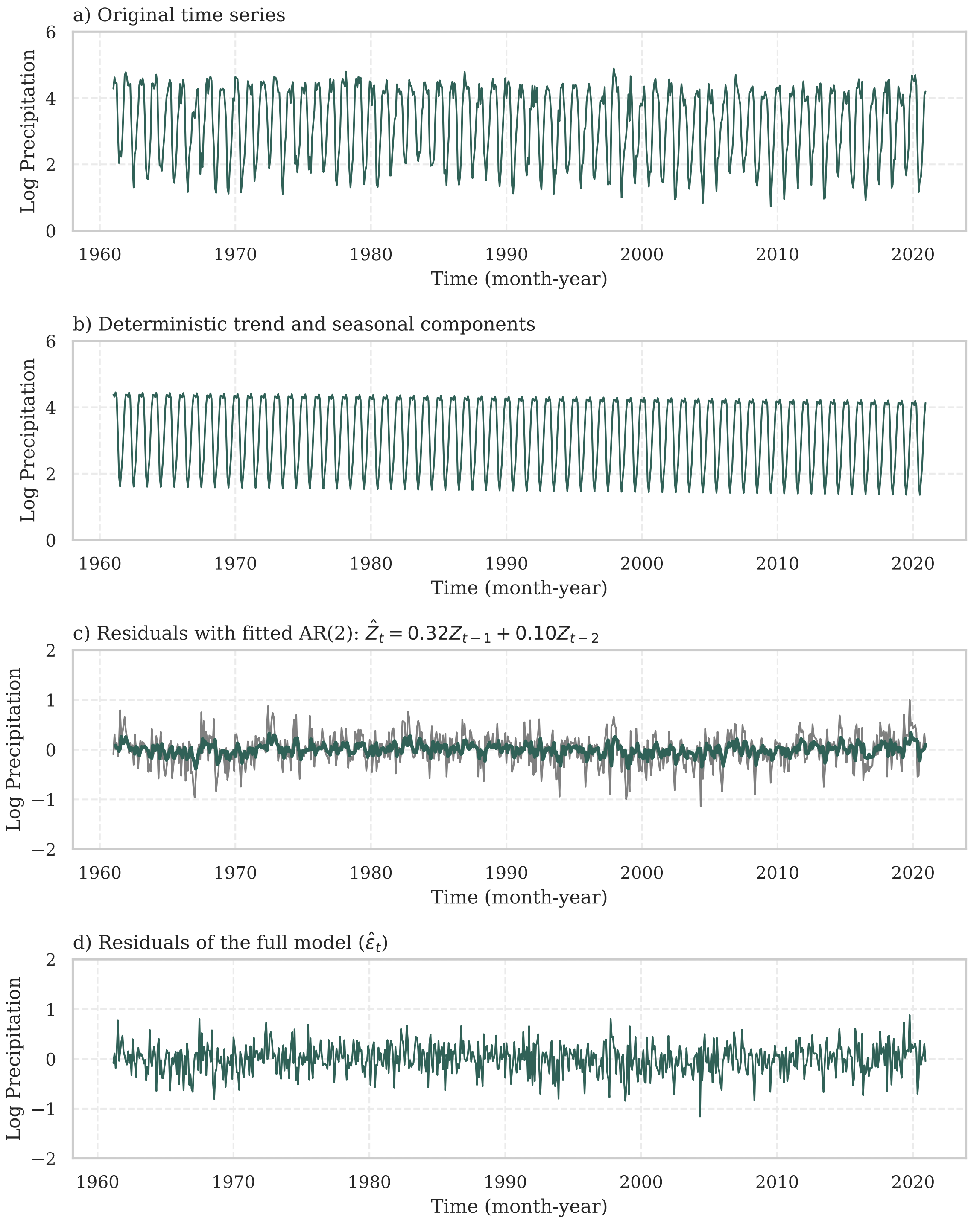

Figure C.6: Log of monthly average precipitation in Tanzania in 1961-2020 with estimated deterministic trend and seasonal components and $AR(2)$ residual. Visualisation of estimation results, GLS restricted model, see Tab. C.4 for detailed results and Fig. C.7 for residual plots (ERA5 data, Author's calculation).

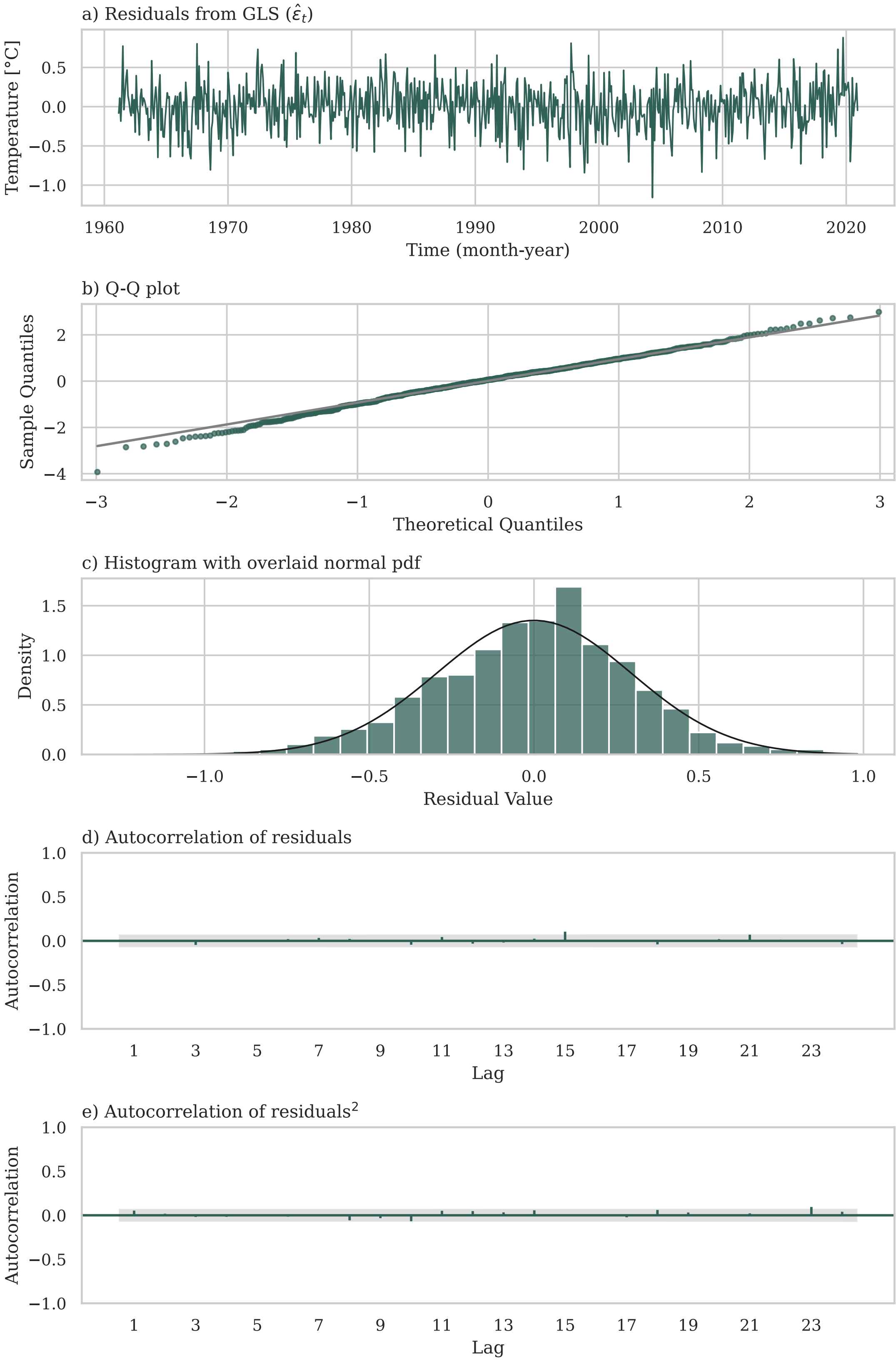

Figure C.7: Log of monthly average precipitation in Tanzania in 1961-2020. Residual plots for GLS restricted model with deterministic trend and seasonal components and $AR(2)$ residual presented in Fig. C.6 above (ERA5 data, Author's calculation).